Inventories and Monetary Policy

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Abstract

We show that the transmission of monetary policy depends sharply on the inventory position of firms. A monetary policy shock affects output and prices more than twice as much when firms hold low inventories. In the high-inventory regime the same shock produces responses that are small and often statistically insignificant. The result holds at the aggregate, sectoral, and firm levels. We rationalize it in a New Keynesian model with stockouts and sticky prices.

JEL classification: E22, E31, E32, E52

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1 Introduction

Inventories are among the most volatile components of aggregate spending, yet their role in monetary policy transmission has received little systematic attention. This paper argues that the oversight is consequential: the inventory position of firms is a first-order determinant of how effectively monetary policy shifts output and prices.

Our main finding is that monetary policy is more than twice as potent when inventories are scarce. Following an unexpected 25-basis-point increase in the one-year Treasury rate, industrial production declines by more than 1.5% at peak, and the PPI by roughly 1%, when firms enter the episode with a low inventory-to-sales ratio. In the high-inventory regime, the corresponding responses are less than half as large. We establish this asymmetry at three levels of aggregation. At the aggregate level, we estimate a Bayesian threshold VAR identified with external instruments, exploiting time-series variation in the business-sector inventory-to-sales ratio to classify the economy into high- and low-inventory regimes. At the sector level, we corroborate the finding using a panel of manufacturing industries, estimating state-dependent local projections that interact the monetary policy shock with a sector-specific inventory indicator. We show that the headline sectorial responses do not materially change when we control for time fixed effects, addressing the concern that, being endogenous to the business cycle, the inventory indicator would capture the differential impact of monetary policy during recessions and expansion, as shown by Tenreyro and Thwaites (2016). At the firm level, we sharpen the identification further by including sector-time fixed effects. We additionally control for several firm-level characteristics that absorb previously documented state-dependent channels of monetary policy—such as firm size (Gertler and Gilchrist, 1994), leverage (Ottonello and Winberry, 2020), and liquidity (Jeenas, 2024). Finally, the asymmetric amplification result is robust across a wide range of choices for the construction of the indicator, the proxy for monetary policy shocks, the sample period, and alternative regression models.

Our findings shed new light on a long-standing puzzle. Standard production-smoothing models predict that firms use inventories to buffer demand fluctuations, insulating output from shocks (Blinder, 1982). The data, however, consistently reject this prediction: production is more volatile than sales, and the inventory-to-sales ratio moves countercyclically (Blinder and Maccini, 1991). Confronted with this failure, the literature has largely concluded that inventories play a negligible role in the aggregate business cycle (Khan and Thomas, 2007), sparing macroeconomists from an additional state variable. Our analysis reframes this conclusion: inventories may not alter unconditional business cycle moments, but they profoundly shape the economy's response to monetary policy.

To rationalize our finding, we embed the stockout-avoidance motive of Kryvtsov and Midrigan (2013) in an otherwise standard New Keynesian model with quadratic costs of price adjustment (Rotemberg, 1982). Firms choose price and production before their own demand is realized, and hold inventories to avoid stocking out. The model attributes the stronger transmission of monetary policy to the response of the stockout probability. When demand falls after a monetary contraction, the stockout probability falls. Since this is also the probability of selling an additional unit of goods, firms demand less labor and cut production. Furthermore, firms' desired price is a markup over the value of a unit kept in stock, and in the model both terms fall after a monetary contraction. The markup falls because fewer sales are truncated by stockouts, making demand more price elastic. The value of a unit kept in stock falls because the real interest rate rises and the unit is less likely to be sold in the future. These effects are stronger when inventories are low, because the stockout probability is then more sensitive to demand. When inventories are low relative to demand, a modest fall in demand makes a stockout much less likely. When inventories are ample, stockouts are already unlikely, so the same demand decline has a much smaller effect.

Our paper contributes to three strands of the literature. The first concerns inventories and the transmission of shocks. An early literature documents that inventory dynamics

are a key conduit for monetary policy: Bernanke and Gertler (1995) find that inventory disinvestment is among the earliest and sharpest responses to a monetary tightening; and Kashyap et al. (1994) show in the cross-section that firms lacking access to public capital markets cut inventories far more sharply during episodes of tight money. de Sousa Rodrigues and Willems (2024) document a cost-of-carry channel: a monetary tightening raises the cost of holding inventories, so firms with larger stocks cut prices to reduce them, and monetary policy lowers goods prices by more when inventories are elevated. In their model the marginal cost of carrying inventories rises with the stock. In our model it does not, and the state dependence comes instead from stockout risk, which is highest when inventories are low. Inventories also shape how demand shocks propagate across firms. Kahn (1987) shows that the motive to avoid stockouts can make production more volatile than sales, and Ferrari (2026) provides causal evidence that final demand shocks are amplified as they travel upstream along supply chains, with inventories raising output elasticities by 17% on average. In Ferrari (2026), amplification comes from firms adjusting their stock in proportion to expected sales, and it rises with the amount of inventory firms hold. Our evidence instead conditions on whether inventories are temporarily low or high, and shows that monetary policy is more potent when they are low.

The second strand concerns stockouts and rationing. A theoretical literature explains inventory behavior through the motive to avoid stockouts (Kahn, 1987). Bils and Kahn (2000) and Bils (2016) use this motive to infer the cyclical behavior of markups, and Kryvtsov and Midrigan (2013) use it to measure how markups and costs respond to monetary shocks. Cavallo and Kryvtsov (2023) construct a direct measure of stockouts from the websites of large retailers in seven countries. They document that about 10% of products were out of stock in the United States before the pandemic, that stockouts rose several-fold early in the pandemic, and that an unexpected increase in a sector's stockout rate raises its inflation within three months, with larger effects in import-intensive sectors. Holden (2025) studies an economy in which firms ration demand when their sticky price falls

below marginal cost. He shows that an expansionary monetary shock raises the news-based shortages index of Caldara et al. (2025), and that rationing makes the Phillips curve convex: expansionary shocks raise rationing, while contractionary shocks reduce it and cushion the fall in output. Our model shares the prediction that a monetary tightening lowers the incidence of stockouts, but with a different implication for quantities. A lower stockout probability lowers the value of the marginal unit of goods, and with it labor demand. Our contribution to this strand is to show that the inventory position governs how much stockout risk responds to monetary policy, and through it how strongly production and prices respond.

The third strand concerns state-dependent supply and pricing responses. Boehm and Pandalai-Nayar (2022) document convex aggregate supply curves, with price responses to demand shocks that are markedly larger when capacity is tight; the industry capacity utilization rate is a sufficient statistic for the supply elasticity. Ahlander et al. (2026) confirm this at the firm level using Swedish producer-price micro data: capacity-constrained firms raise prices substantially more in response to an expansionary monetary shock, with the difference widening when competitors are similarly constrained. Bai et al. (2026) apply a similar logic to global supply chain disruptions: using a structural threshold VAR and local projections, they find that during disruption episodes, when logistical bottlenecks steepen the aggregate supply curve, contractionary monetary policy produces a larger price decline at a smaller output cost. Mehrotra et al. (2025) embed the inventory-to-sales ratio directly in a New Keynesian Phillips curve and show that it helps explain both the missing disinflation of 2009–2011 and the COVID-era inflation surge through time-varying retail markups. Our contribution to this strand is to show that the inventory position shapes the transmission of monetary policy to quantities as well as to prices.[1]

[1] Our paper also relates to the broader literature on state-dependent monetary policy transmission across financial conditions (e.g., Alessandri and Mumtaz, 2017, 2019). We complement this literature by identifying a real-side amplifier, the inventory position, that is distinct from financial frictions and operates through firms' risk of stocking out rather than their access to external finance.

The rest of the paper proceeds as follows. Section 2 presents the aggregate evidence from the threshold VAR. Section 3 turns to a panel of manufacturing industries, where time fixed effects absorb the aggregate business cycle, and Section 4 to firm-level data, where industry-by-quarter fixed effects absorb every aggregate and industry-wide confounder. Section 5 develops the model. Section 6 concludes.

2 Aggregate evidence

We begin at the aggregate level, where firms' inventory position is summarized by the economy-wide inventory-to-sales ratio. We document the state dependence of monetary policy using two complementary strategies: a Bayesian threshold VAR that sorts the economy into high- and low-inventory regimes, and state-dependent local projections that relax its parametric structure. Both deliver the same conclusion: monetary policy is markedly more effective when inventories are low.

2.1 Threshold VAR

There are several possible approaches to estimating the state-dependent effects of a monetary policy shock as a function of the amount of inventories. As a first step, we follow Alessandri and Mumtaz (2017, 2019) and estimate a Bayesian threshold VAR. We start with this model because it allows us to estimate the state-dependent effects of a monetary policy shock through generalized impulse response functions (GIRFs). As argued by Francis et al. (2023) and Gonçalves et al. (2024), GIRFs account for endogenous transitions across states induced by the shock, thereby avoiding the potential bias of local projections (LPs), which instead capture average effects across regimes. Moreover, the reduced-form VAR coefficients are estimated in a single step, improving overall model fit.

Model. Consider the following threshold VAR model:

$$Y_t = \left[c_1 + \sum_{j=1}^{P} B_{1,j} Y_{t-j} + u_{1,t} \right] S_t + \left[c_2 + \sum_{j=1}^{P} B_{2,j} Y_{t-j} + u_{2,t} \right] (1 - S_t) \tag{1}$$

where $Y_t = [y_{1,t},\ y_{2,t},\ \cdots,\ y_{n,t}]'$ is a $n \times 1$ vector of endogenous variables; $u_{i,t} \sim \mathcal{N}(0, \Sigma_i)$ is a $n \times 1$ vector of reduced-form innovations in regime $i \in \{1, 2\}$, with Σ_i the associated $n \times n$ covariance matrix; and c_i and $B_{i,1}, \ldots, B_{i,p}$ denote, respectively, the $n \times 1$ vector of intercepts and the $n \times n$ matrices of autoregressive slopes in regime $i = 1, 2$. The regime indicator S_t is defined as:

$$S_t = 1 \iff y_{n,t-d} \leq y_n^*, \tag{2}$$

where $y_{n,t}$ is the threshold variable, assumed without loss of generality to be the last element of Y_t, y_n^* is the threshold value, and $d \geq 1$ is the delay parameter governing the minimum lag with which $y_{n,t}$ can affect regime classification. Accordingly, $S_t = 1$ identifies the "low" regime, prevailing when the threshold variable lies below y_n^*, while $S_t = 0$ identifies the "high" regime.

Estimation. The unknown parameters $\boldsymbol{\theta} \equiv \{B_{i,j},\ c_i,\ \Sigma_i\}_{i=1,2;\ j=1,\ldots,p}$, together with the threshold value y_n^* and the delay parameter d, are estimated by Bayesian methods. We adopt a Minnesota prior, which shrinks the VAR slope matrices toward univariate AR(1) processes and toward zero at longer lags, complemented by a sum-of-coefficients prior that further pulls the sum of lag coefficients toward a unit root. Importantly, both priors are applied symmetrically across regimes, implying a conservative approach: the two states share a common prior that shrinks their coefficients toward being identical, so that differences across regimes emerge only to the extent warranted by the data.

The joint posterior distribution does not admit a closed form and is simulated via a Markov chain Monte Carlo (MCMC) algorithm that combines a Gibbs sampler for $\boldsymbol{\theta}$ with a Metropolis-Hastings step for the threshold y_n^* and a discrete Gibbs step for the delay d. Further details on the estimation procedure are provided in Appendix B.1.

Variables and sample. The VAR is estimated at a monthly frequency and includes six variables. The 1-year Treasury bill rate, in basis points, is used to normalize the impulse responses to a monetary policy shock. The industrial production index (IP) and the pro-

ducer price index (PPI), both in logs, capture real activity and nominal dynamics, respectively. The log-transformation of commodity prices and the excess bond premium (EBP, in basis points), both highly forward-looking, serve as controls for financial conditions. The last variable, which serves as the threshold variable, is the inventory-to-sales ratio, whose specification is discussed below.

The numerator of the threshold variable is finished-goods inventories of the business sector.[2] The denominator is the cumulative sales over the current and preceding two months, approximating the last quarter's total sales rather than current-month sales alone. This allows for a slow-moving normalization of the available stock of inventories, so that variations in the ratio are primarily driven by changes in the numerator rather than by high-frequency swings in sales that partly reflect variation in aggregate demand. Moreover, since the inventory-to-sales ratio exhibits pronounced low-frequency trends, we apply a high-pass filter that removes fluctuations with frequencies lower than ten years.[3] Details on data sources are provided in Appendix A.

Finally, the VAR is estimated with $p = 12$ lags, consistent with the monthly frequency of the data and the sample runs from December 1981—the earliest date at which business-sector final-goods inventories are jointly available across manufacturing, wholesale, and retail—to January 2020, which excludes the COVID-19 period.

Inventory regimes. Figure 1 displays the estimated regime classification. The black line is the filtered inventory-to-sales ratio $y^*_{n,t-d}$ evaluated at the posterior mode of the delay parameter d, the red horizontal line is the posterior median of the estimated threshold y^*_n, and the shaded vertical areas identify periods in the low-inventory regime when using the posterior mode for d and the posterior median for y^*_n.

[2] We focus on output rather than input inventories because finished-goods stocks directly measure firms' capacity to fulfill demand. A firm may hold large input inventories yet remain unable to supply if even one key input is missing, whereas final-goods inventories unambiguously signal that finished goods are available for sale.

[3] The results are robust to measuring the cyclical component with the regression filter of Hamilton (2018), which is one-sided.

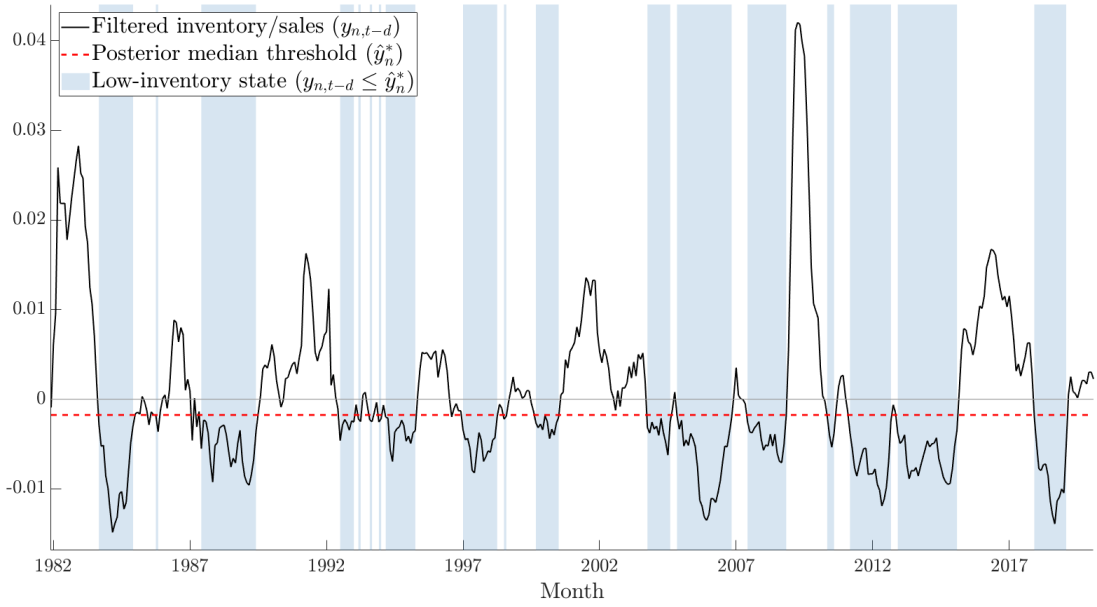


Figure 1: Filtered inventory-to-sales ratio and regime classification

Note: The black line is the cyclical component of the finished-goods inventory-to-sales ratio of the business sector, obtained by applying a high-pass filter that removes fluctuations with frequencies lower than 120 months, and evaluated at the posterior mode of the delay parameter d. The red horizontal line is the posterior median of the estimated threshold y_n^*. Shaded areas identify periods classified as the low-inventory regime ($S_t = 1$), defined using the posterior mode of d and the posterior median of y_n^* to assign each observation to a regime. Sample: December 1981–January 2020. See Appendix B.1 for a detailed description of the estimation of the reduced-form VAR.

The estimated threshold lies slightly above zero, reflecting the fact that the posterior does not simply coincide with the average of the state variable $y_{t,n}$ (which is zero by the property of the filter) but is also shaped by the likelihood of the data. Moreover, the regime classification does not simply track the business cycle: while some low-inventory episodes coincide with recessions, the figure reveals variation that is independent of aggregate downturns, suggesting that inventory positions may contain information about monetary policy transmission beyond what is captured by standard measures of economic slack.

Identification. Following Stock and Watson (2018), we identify the causal effect of monetary policy by augmenting the reduced-form threshold VAR with an external instrument, using existing off-the-shelf proxies for the monetary policy shock series. The key addi-

tional step relative to a standard external-instrument VAR is that the first-stage regression is estimated separately for each regime. Specifically, let $\hat{u}_{i,t}$ denote the estimated vector of reduced-form residuals in regime $i \in \{1, 2\}$. For each state, we regress $\hat{u}_{i,t}$ on the proxy for the monetary policy shock $\hat{\varepsilon}_t$:

$$\hat{u}_{i,t} = \gamma_i \, \hat{\varepsilon}_t + v_{i,t}, \qquad i = 1, 2, \tag{3}$$

where under conventional assumptions the estimated coefficient vector $\hat{\gamma}_i \in \mathbb{R}^n$ captures the impact effect of the monetary policy shock on the endogenous variables Y_t in regime i. The baseline proxy is the monetary policy surprise series of Bauer and Swanson (2023), which purges high-frequency interest rate surprises of their predictable component using publicly available macroeconomic information, and represents one of the most recent advances in the measurement of monetary policy shocks. As discussed in Section 2.3, the results are robust to the use of alternative monetary policy proxies.

This approach offers two key advantages. First, the reduced-form stage and the structural stage are independent and can be estimated separately. This separation property is particularly valuable when studying monetary policy transmission, since monetary policy shock proxies are available only over a restricted time window relative to the availability of standard macroeconomic indicators. Thanks to this separation, we can estimate the reduced-form VAR parameters over the full sample for which all endogenous variables are available, and then identify the state-specific impact vectors $\hat{\gamma}_i$ on the shorter overlapping sample between the reduced-form residuals and the monetary policy proxy $\hat{\varepsilon}_t$. Second, the threshold VAR framework allows us to construct state-dependent generalized impulse response functions (GIRFs; Koop et al. 1996). State-dependent GIRFs condition on the regime prevailing at the time the shock occurs and allow the economy to transition between regimes endogenously after impact. As argued by Francis et al. (2023) and Gonçalves et al. (2024), this avoids the potential bias of local projections, which instead

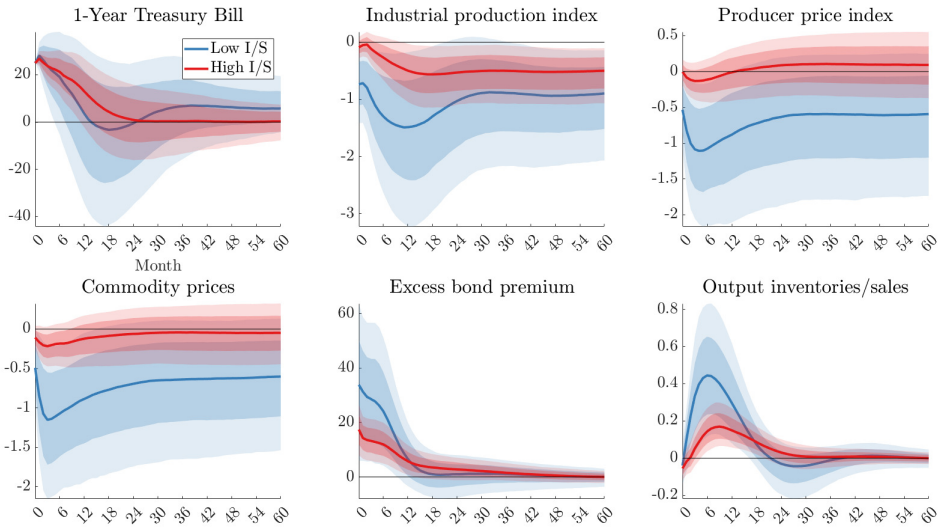
capture average effects across regimes. We describe the construction of the GIRFs in Appendix B.2.

2.2 Responses by inventory regime

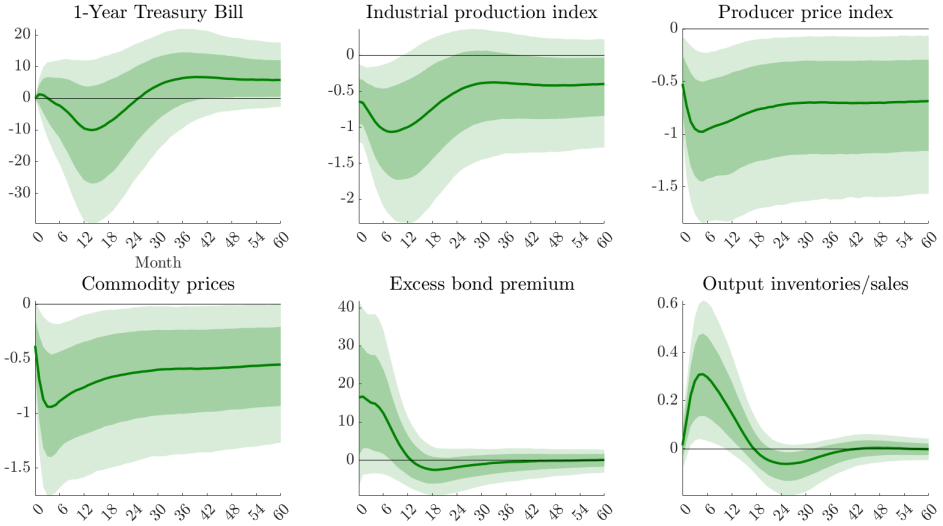
Figure 2 presents the GIRFs to a contractionary monetary policy shock for all variables in the VAR (Panel a) together with the corresponding differences across regimes (Panel b). The shock is normalized so that the impact effect on the one-year Treasury bill equals 25 basis points. Solid lines are posterior medians; dark and light shaded areas correspond to 68% and 90% posterior credible bands, respectively. Blue (red) lines and bands correspond to the low- (high-) inventory regime, defined as periods when the inventory-to-sales ratio lies below (above) the estimated threshold y_n^*.

Considering first the one-year Treasury bill rate, the response decays quickly and becomes statistically non-significant after approximately one year, with broadly similar dynamics across both regimes. Turning to industrial production, the producer price index, and commodity prices, we find a sharp contrast across regimes. In the high-inventory regime, the response of industrial production is relatively small, with a peak fall of approximately 0.5%, while the price responses are essentially zero and statistically insignificant throughout. In the low-inventory regime, responses are considerably more pronounced: industrial production declines by approximately 1.5% at peak, while prices fall by about 1%. As shown in Panel (b) of Figure 2, these differences are statistically significant at the 90% level. Moreover, both the excess bond premium and the inventory-to-sales ratio confirm their countercyclical nature, displaying a positive response in both regimes, with these responses again amplified in the low-inventory regime.

Results reveal a sharp asymmetry in monetary policy transmission. A tightening that occurs when firms hold high inventories appears nearly ineffective: responses in economic activity and prices are either small or statistically insignificant. By contrast, when firms enter the contractionary episode with low inventories, the same shock generates a pronounced and statistically significant contraction in industrial production and prices.



(a) State-dependent GIRFs to a monetary policy shock



(b) Difference in state-dependent GIRFs

Figure 2: Baseline specification

Note: Panel (a) shows the state-dependent GIRFs; Panel (b) shows the difference between the low- and the high-inventory regime responses. Each panel reports the response of the endogenous variables to a monetary policy shock, normalized so that the impact effect on the Treasury bill equals 25bp. Response of the Treasury bill and EBP are in basis point, while the response of the remain variables are in percent. Solid lines are posterior medians; dark and light shaded areas correspond to 68% and 90% posterior credible bands, respectively; blue (red) lines and bands correspond to the low- (high-) inventory regime. The state-dependent impact effect γ_i of the shock is identified using the proxy of Bauer and Swanson (2023).

2.3 Robustness

The baseline specification presented above involves several choices that, while defensible on econometric grounds, are ultimately arbitrary. To assess whether our findings are an artifact of these choices, we subject the baseline to a battery of perturbations. In each case, we modify one dimension of the specification at a time and verify that the asymmetric amplification result survives. For brevity, all results are relegated to Appendix E; here we describe the perturbations considered and summarize the main findings.

Monetary policy shock series. Our baseline proxy for the monetary policy shock series follows Bauer and Swanson (2023). While this proxy represents one of the most rigorous recent advances, the literature offers several complementary approaches. We therefore re-estimate the model using three alternative instruments. First, we consider the series of Miranda-Agrippino and Ricco (2021) as extended by Degasperi and Ricco (2024), which removes the information component from monetary policy announcements, isolating a purer policy surprise. Second, we use the proxy of Romer and Romer (2004) as updated by Wieland (2021), identified via a narrative approach. Third, we use the series of Aruoba and Drechsel (2024), which extracts monetary policy shocks from Federal Reserve communications via a large language model.

See Figures E2, E3, and E4 of the Appendix for the results. The asymmetric amplification result is qualitatively robust across all three alternatives: both industrial production and prices display the same direction of asymmetry as in the baseline, with the low-inventory regime consistently showing a larger contraction than the high-inventory regime. Statistically, the response of industrial production remains strongly significant, similar to the baseline, while the price responses are somewhat less precisely estimated. Quantitatively, the magnitude of the effects is somewhat smaller than in the baseline for both variables. All in all, these findings reinforce the conclusion that the low-inventory state amplifies

monetary policy transmission, with the asymmetry in real activity being the most robustly estimated dimension.

Inventory measure. The baseline inventory-to-sales ratio uses finished-goods inventories as the numerator. While this is the most direct measure of goods available for sale, we assess robustness along three dimensions. First, we subtract backlogs of orders from the numerator, so that the state variable captures inventories of goods immediately ready for delivery rather than production already committed to future orders. Second, we consider two alternative inventory aggregates: input inventories (raw materials and work-in-progress in manufacturing) and total inventories (the sum of all types of inventories), which, while not the exact object of interest, provide broader or upstream measures of firms' inventory positions. Third, to address concerns that the regime classification may reflect contemporaneous demand conditions rather than firms' pre-existing inventory positions, we replace the last-quarter sales denominator with the last two quarters and over the last year, respectively.

See Appendix Figures E5, E6, E7, E8, and E9 for the estimated differences in state-dependent GIRFs under each alternative. The asymmetric amplification result holds across all the cases: contractionary monetary policy consistently generates statistically significant larger contractions in the low-inventory regime. Effects on prices are in some cases somewhat less pronounced than in the baseline.

Threshold delay and classification rule. Two aspects of the regime-classification scheme are assessed. First, the delay d entering the threshold condition $y_{n,t-d} \leq y_n^*$ is selected endogenously by the discrete Gibbs step of the MCMC, with the baseline posterior concentrating on $d = 2$. We verify that the asymmetric amplification result is not an artifact of this particular value by fixing $d = 1$ and $d = 3$ in turn. Second, in the baseline y_n^* is treated as an unknown parameter and estimated jointly with the rest of the model via the Metropolis–Hastings step, with a prior centered at zero. We additionally consider a non-Bayesian classification rule that fixes $y_n^* = 0$, which equals the unconditional mean of the

band-pass-filtered IS ratio by construction, classifying regimes deterministically according to whether the state variable lies above or below its long-run trend. The asymmetric amplification result survives in all cases.

Additional variables. The baseline VAR includes the producer price index (PPI) as a nominal variable. To assess whether the results extend to alternative price and quantity measures, we consider two extensions. First, we replace the PPI with the consumer price index (CPI), which more directly captures the cost of living for a representative household. Second, we add the unemployment rate as an additional endogenous variable to verify that the responses of industrial production are mirrored in the labor market.

See Figures E10 and E11 of the Appendix for the results. The CPI response is qualitatively consistent with the PPI, confirming the asymmetric amplification pattern, though less pronounced. The unemployment rate shows a difference in response between the two regimes of almost 15 basis points, consistent with the industrial production dynamics.

2.4 State-dependent local projections

A further question is whether the findings are robust under alternative empirical approaches. VAR models impose a particular structure on the economy, and inference may be distorted when the data-generating process departs from that structure. Local projections (Jordà, 2005) provide a complementary approach by estimating the response of each outcome directly at every horizon, thereby relaxing some of the parametric restrictions imposed by the VAR. Although local projections generally entail greater sampling uncertainty (Plagborg-Møller and Wolf, 2021) and may be biased in state-dependent settings (Gonçalves et al., 2024), they provide an informative comparison. Qualitatively similar responses would suggest that the main findings are not driven by the particular estimation framework of the threshold VAR.

We implement the local-projection estimator of Jordà (2005), also formulated in a Bayesian framework by Miranda-Agrippino and Ricco (2021), and extend it to allow for

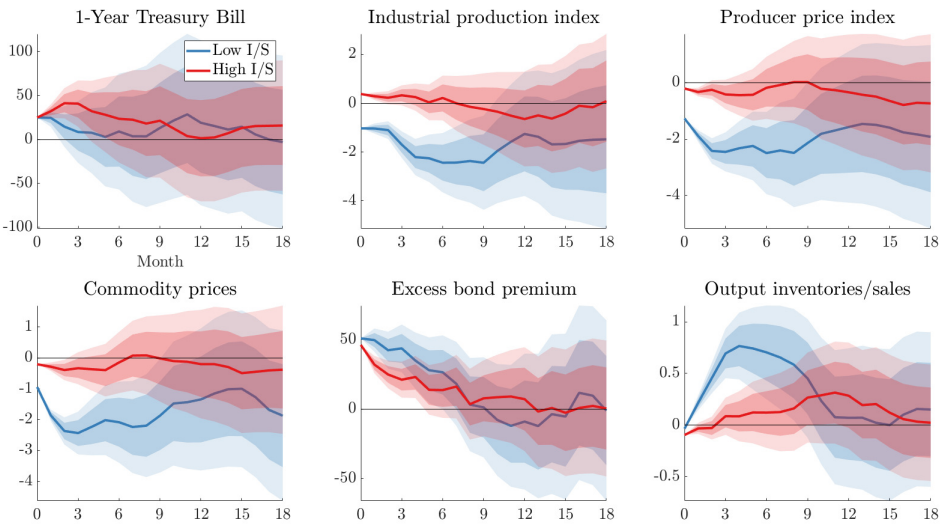
state dependence.[4] The specification includes the same endogenous variables and controls as equation (1), with 12 lags as in the baseline VAR. We construct the threshold variable exactly as in the baseline specification, but impose the regime classification rather than estimating it. Specifically, $S_t = 1$ if $y_{n,t-1}$ is below trend, and zero otherwise, where $y_{n,t}$ denotes the filtered inventory-to-sales ratio used in the baseline VAR. Inference is based on Bartlett-kernel Newey–West heteroskedasticity- and autocorrelation-consistent standard errors. Appendix C provides further details on estimation and inference.

Figure 3 presents the state-dependent impulse response functions of all variables in the baseline specification to a contractionary monetary policy shock, together with the implied differences in responses across regimes. Solid lines are point estimates; dark and light shaded areas correspond to 68 and 90 percent confidence bands, respectively; blue and red lines and bands correspond to the low- and high-inventory regimes. The results are consistent with the baseline specification. Across variables, monetary policy has a stronger effect in the low-inventory regime, and the differences across regimes are statistically significant and quantitatively similar to those implied by the VAR.[5]

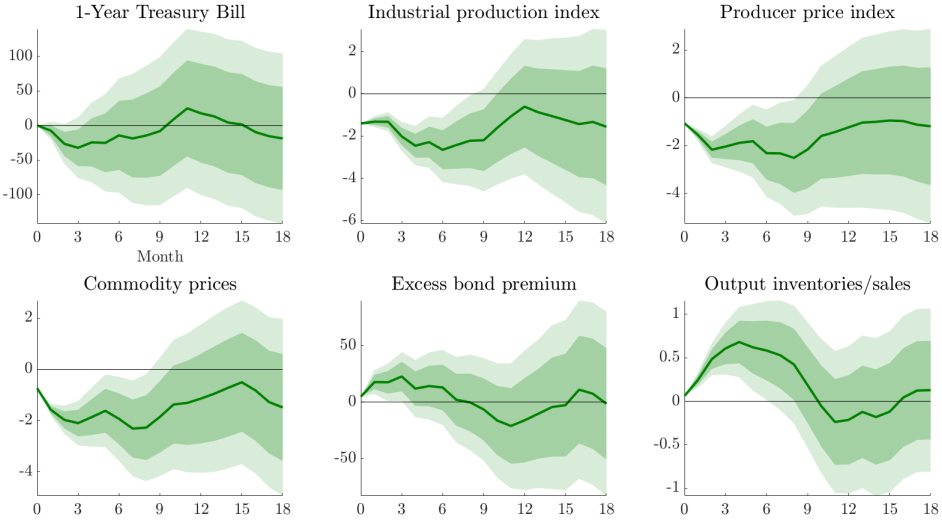
Overall, the fact that a different estimator continue to deliver qualitatively consistent asymmetric amplification provides strong confirmation of the baseline results. These exercises also serve as a natural bridge to the next section, where we employ local projections to show that the state-dependent amplification is not limited to time variation alone but survives when identification exploits cross-sectional variation across sectors.

[4] We use this approach rather than a direct regression of the endogenous variables on the shock of interest, as the latter would restrict the entire estimation to the shorter period over which the Bauer and Swanson (2023) proxy is available. Separating reduced-form dynamics from structural identification allows us to estimate the former on the full sample and use the shorter overlapping sample only to recover the state-specific structural impact vectors.

[5] Note that we display responses through 18 months, since beyond that horizon the confidence bands widen considerably and obscure the shorter-run dynamics. For completeness, Figure D4 in Appendix D reports the responses over the full five-year horizon.



(a) State-dependent IRFs to a monetary policy shock



(b) Difference in state-dependent IRFs

Figure 3: Local projections

Note: Panel (a) shows the state-dependent IRFs; Panel (b) shows the difference between the low- and the high-inventory regime responses. Each panel reports the response of the endogenous variables to a monetary policy shock, normalized so that the impact effect on the Treasury bill equals 25bp. Response of the Treasury bill and EBP are in basis point, while the response of the remain variables are in percent. Solid lines are point estimates; dark and light shaded areas correspond to 68% and 90% confidence bands, respectively; blue (red) lines and bands correspond to the low- (high-) inventory regime. The effects of the shock are identified using the proxy of Bauer and Swanson (2023).

3 Sector-level evidence

Because the *aggregate* inventory-to-sales ratio co-moves with economic activity, the asymmetric transmission documented above may partly capture other macroeconomic factors. We therefore turn to sector-level data, exploiting more granular variation to isolate how inventory positions shape the effects of monetary policy.[6]

3.1 Data

The sector analysis combines monthly data from three sources. We source inventories and sales from the U.S. Census Bureau's Manufacturers' Shipments, Inventories, and Orders (M3) survey, which covers U.S. manufacturing industries from 1992 onward. We merge these data with the industrial production from the Federal Reserve Board (G17) and producer prices indexes from the Bureau of Labor Statistics (BLS), which we use as our measures of real activity and prices, respectively. Because industrial production indexes are available for manufacturing industries but, by construction, not for wholesale or retail industries, we restrict the analysis to manufacturing. Our final sample comprises 21 three-digit NAICS manufacturing industries and runs from January 1992 through December 2023. Appendix A.2 provides further details on data construction and sample coverage.

For each sector, we construct the inventory-to-sales ratio using the same methodology as in the aggregate analysis. The numerator denotes finished-goods inventories while the denominator is the cumulative sales over the current and preceding two months. We then apply a high-pass filter that removes fluctuations with frequencies lower than ten years. A sector is therefore classified as being in the low-inventory regime if its lagged inventory-to-sales measure is below zero. Importantly, Figure 8 in Appendix A.2 documents substantial cross-sectional variation in sectors' filtered inventory-to-sales ratios within each period.

[6] A non-exhaustive list of aggregate conditions shown to shape monetary policy transmission includes the state of the business cycle (Tenreyro and Thwaites, 2016); financial stress (Dahlhaus, 2017); and economic uncertainty (Aastveit et al., 2017).

This variation allows us to identify the state-dependent effects of monetary policy while including time fixed effects that absorb potential aggregate confounding factors.[7]

3.2 Specification

To investigate whether the effects of monetary policy depend on sectors' inventory positions, we estimate variations of the following state-dependent local projection:

$$y_{j,t+h} - y_{j,t-1} = \beta_h \varepsilon_t + \delta_h \left(\hat{\varepsilon}_t \times S_{j,t-1}\right) + \gamma_h X + \alpha_{j,h} + \alpha_{t,h} + u_{j,t+h}, \tag{4}$$

where $y_{j,t}$ denotes one of the dependent variables: the log of the sector-level industrial production index, producer price index, or real sales, or the unfiltered inventory-to-sales ratio in levels. As in the aggregate analysis, $S_{j,t-1}$ indicates whether sector j is in a low inventory state in period $t-1$, and $\hat{\varepsilon}_t$ is the monetary policy shock proxy from Bauer and Swanson (2023), normalized to raise the one-year Treasury rate by 25 basis points on impact over our sample. The control vector X includes 12 lags of the endogenous variables used in the aggregate analysis, the proxy $\hat{\varepsilon}_t$, and the first difference of the dependent variable, together with the interactions of these variables with $S_{j,t-1}$. Intercepts $\alpha_{j,h}$ and $\alpha_{t,h}$ represent sector and time fixed effects, respectively. Because 21 clusters across sectors are insufficient to obtain reliable standard errors, we use Driscoll–Kraay standard errors to account for correlations across sectors and time.

3.3 Responses by inventory state

We begin by estimating a version of equation (4) that omits the time fixed effects, $\alpha_{t,h}$. The advantage of this less saturated specification is that it allows to recover the level impulse response in each inventory regime. The cost is that it absorbs only time-invariant differences across sectors, leaving aggregate movements common to all sectors as part of the identifying variation.

[7] Time fixed effects account for only about 20% of the variation in the sectoral inventory-state indicator.

Figure 4 reports the impulse responses of the endogenous variables across inventory regimes. Responses in the low-inventory regime, captured by $\beta_h + \delta_h$, are shown by the solid blue line, whereas those in the high-inventory regime, captured by β_h, are depicted by the dashed red line. The corresponding shaded areas delimit the 90% confidence bands. Results are consistent with the state-dependent pattern of the aggregate analysis. A contractionary monetary policy shock, normalized to an increase of 25 basis points in the 1-year Treasury rate, reduces production, real sales, and prices in both inventory regimes, with more pronounced effects when a sector is in a low-inventory regime. The response of industrial production, for instance, falls by 9.3 percent in the low-inventory state, compared with 3.5 percent in the high-inventory one. The inventory-to-sales ratio rises more strongly in the low-inventory regime, indicating that inventories adjust proportionally less than sales.[8]

Figure 5 reports the differences in the impulse responses of the endogenous variables across inventory regimes, as captured by δ_h. The solid green line corresponds to the differences implied by the responses shown in Figure 4, while the dashed line presents estimates from the most saturated specification of equation (4), which includes time fixed effects. The correspondingly shaded areas denote the 90% confidence bands.

In the specification without time fixed effects, the differences visible in Figure 4 are statistically significant, confirming that monetary policy transmission varies across inventory regimes. More importantly, the specification with time fixed effects allows to assess whether the asymmetry documented thus far reflects the stock of inventories available or other underlying aggregate factors. By absorbing all time variation common to the sectors, time fixed effects isolate the identifying variation in δ_h to changes in inventory positions across sectors over time. This specification therefore allows us to disentangle the role of

[8] Although the sector-level estimates are qualitatively consistent with the aggregate results, the implied responses are larger in magnitude. Two factors may account for this difference. First, the sector-level analysis focuses on manufacturing, which is substantially more volatile than the economy as a whole. Second, because the analysis assigns equal weight to each disaggregated industry, smaller—and potentially more volatile—industries exert the same influence on the estimates as larger ones.

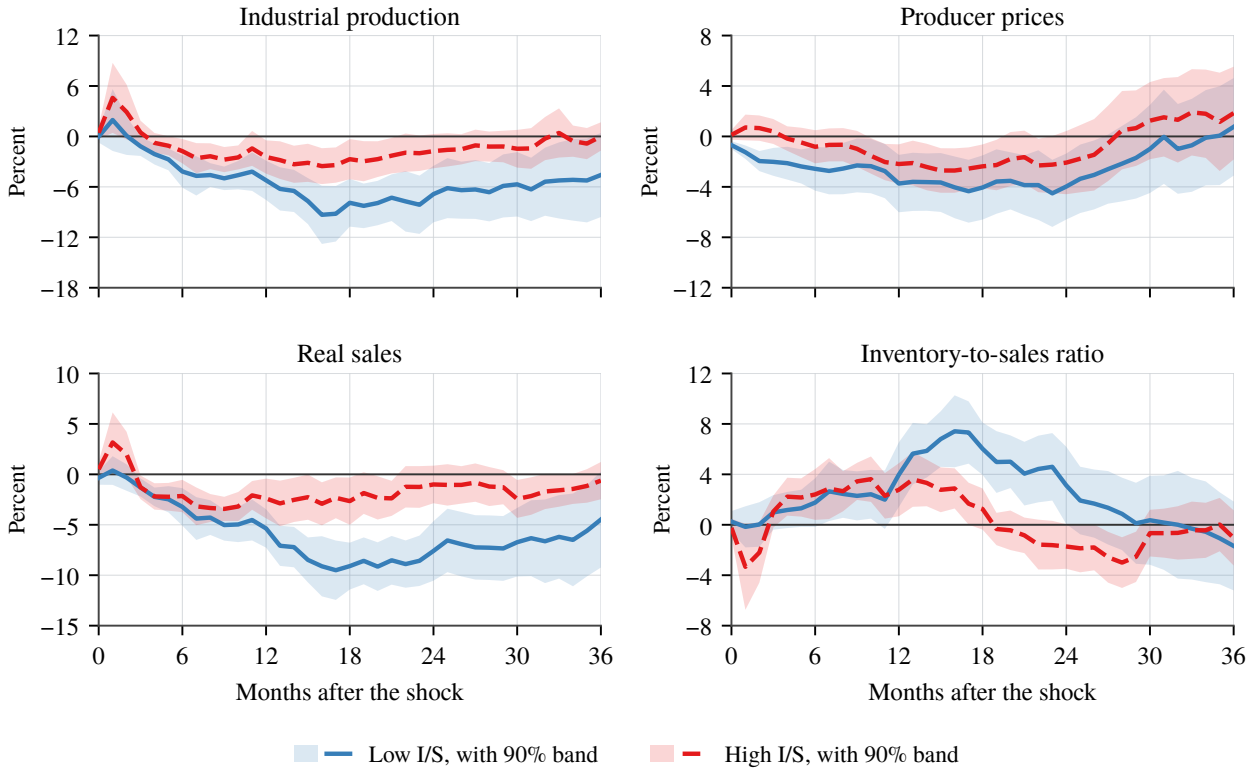


Figure 4: Impulse responses to a monetary policy shock

Note: Average responses of the high-inventory (β_h) vs. low-inventory ($\beta_h + \delta_h$) sectors from Equation (4) with sector fixed effects. Real sales are deflated by the sector's own producer price index. Shaded areas are 90% confidence intervals based on Driscoll–Kraay standard errors to account for correlations across sectors and time, where the maximum lag order is set to the maximum projection horizon. Bauer and Swanson monetary policy shocks are normalized to raise the one-year Treasury rate by 25 basis points on impact.

inventories from that of aggregate confounding factors in shaping monetary policy transmission. The fact that the estimates remain highly significant and quantitatively similar to those obtained without time fixed effects substantially alleviates these concerns.

4 Firm-level evidence

The aggregate and sectoral exercises have two main shortcomings. First, the inventory state is defined at a relatively coarse level of aggregation, thus potentially contaminated by compositional changes within sectors. Second, the inventory-to-sales ratio is subject to filtering, which may induce spurious correlations. To address these concerns, we move to a finer level of disaggregation and examine responses at the firm level. By comparing

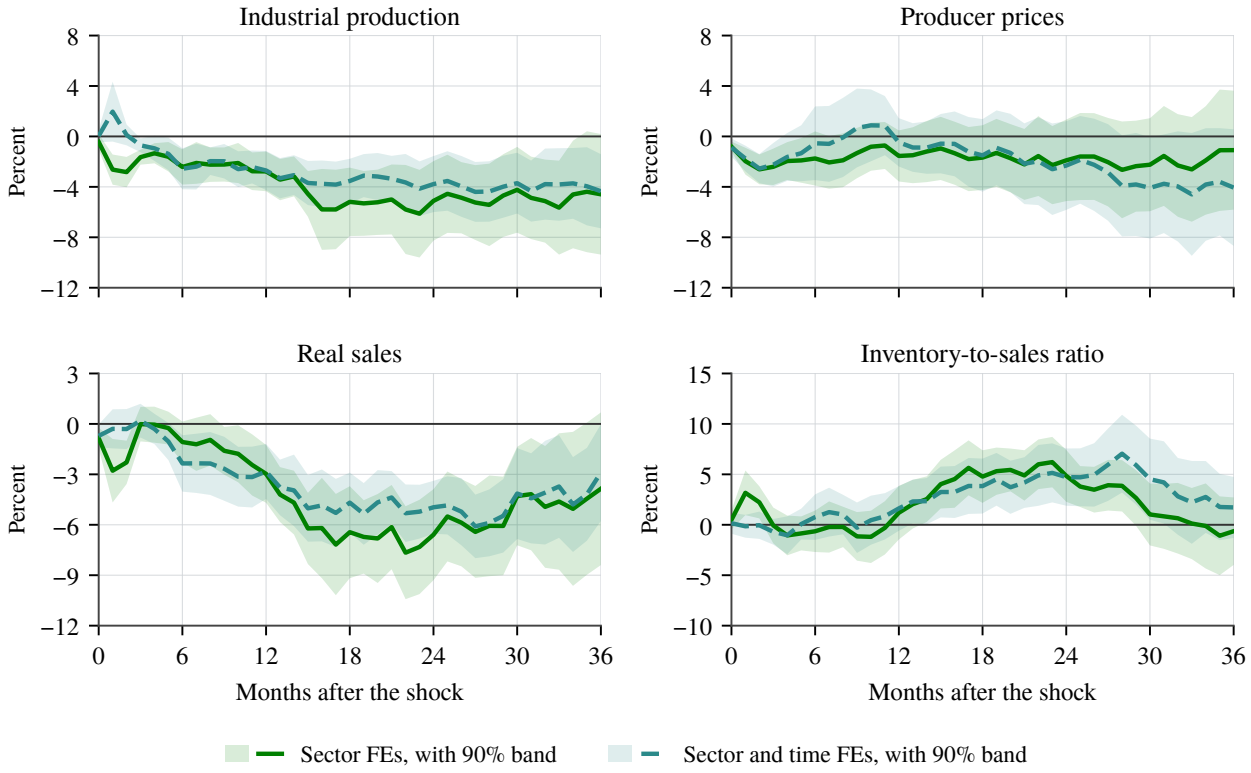


Figure 5: Heterogeneous effects of monetary policy shocks

Note: Differential response of the low-inventory sectors relative to the high-inventory sectors (δ_h) from Equation (4) with and without time fixed effects. Real sales are deflated by the sector's own producer price index. Shaded areas are 90% confidence intervals based on Driscoll–Kraay standard errors to account for correlations across sectors and time, where the maximum lag order is set to the maximum projection horizon. Bauer and Swanson monetary policy shocks are normalized to raise the one-year Treasury rate by 25 basis points on impact.

firms within the same industry and quarter, we exploit the deviation of a firm's inventory position from that of its industry. This approach allows us to control for heterogeneous firm-level responses to monetary policy shocks, as documented in (Jeenas, 2024; Ottonello and Winberry, 2020; Cloyne et al., 2023), while avoiding the need to filter the inventory-to-sales ratio.

4.1 Data and inventory state

We use quarterly Compustat data on U.S. firms outside the financial, utility, and quasi-governmental sectors over 1988Q4–2020Q4. Following Cloyne et al. (2023), we drop firm-quarters with acquisitions exceeding five percent of total assets, trim the key variables at

the first and ninety-ninth percentiles within each year, and keep firms with at least twenty quarters of observations.[9] We retain only firm-quarters for which the inventory-to-sales ratio, tangible capital, and every firm-level control are observed, resulting in a sample of $3{,}815$ firms and $195{,}052$ firm-quarters.[10]

We measure a firm's inventory position by its total inventory-to-sales ratio and classify firm j as being in the *low-inventory state* in quarter t whenever this ratio lies below the median ratio of its three-digit industry in that quarter:

$$S_{j,t} \;=\; \mathbb{1}\Big\{\big(\tfrac{\text{inventories}}{\text{sales}}\big)_{j,t} \;<\; \operatorname{med}_{k\in s(j)}\big(\tfrac{\text{inventories}}{\text{sales}}\big)_{k,t}\Big\}, \tag{5}$$

where $s(j)$ denotes the three-digit SIC industry of firm j, and a firm lying above that median is in the high-inventory state. Firm-quarters whose ratio equals the industry median exactly are left unclassified and do not enter the estimation.

4.2 Specification

We estimate the state-dependent local projection

$$k_{j,t+h} - k_{j,t-1} \;=\; \delta_h\,(\hat{\varepsilon}_t \times S_{j,t-1}) \;+\; \gamma_h X \;+\; \alpha_{j,h} \;+\; \alpha_{s(j),t,h} \;+\; \varepsilon_{j,t+h}, \tag{6}$$

where $k_{j,t}$ is the log of firm j's tangible capital, so the left-hand side is the cumulative change in (log) capital between $t-1$ and $t+h$. $\hat{\varepsilon}_t$ is the Bauer and Swanson (2023) monetary policy shock and $S_{j,t-1}$ is the lagged indicator from (5). The term $\alpha_{j,h}$ is a firm fixed effect and $\alpha_{s(j),t,h}$ is an industry$\times$quarter fixed effect. As in the sector-level analysis, δ_h is the differential response of low-inventory firms relative to high-inventory ones. Standard

[9] We drop the top percentile of the inventory-to-sales ratio, log total assets, book leverage, the quick ratio, Tobin's Q, total inventories, quarterly inventory growth, log sales, and the investment rate, and the bottom percentile of the same variables except book leverage and the quick ratio, which we allow to equal zero following Ottonello and Winberry (2020) and Jeenas (2024).

[10] Requiring a non-missing distance to default costs roughly thirty percent of the observations, a shrinkage comparable to the one reported by Ottonello and Winberry (2020) and Jeenas (2024). We nevertheless verified that results do not materially change if we avoid the loss of observations by excluding distance to default from the control set.

errors are two-way clustered by firm and by quarter. Throughout, we normalize the shock so that its impact effect on the one-year Treasury rate equals 25 basis points over the span for which the shock and our controls are jointly available.

4.3 Results

Figure 6 reports results for two specifications: one with a parsimonious set of controls and one with an expanded set. In Panel (a), the control vector X includes the standalone indicator $S_{j,t-1}$, four lags of each aggregate variable used in the time-series analysis—namely, the growth rates of industrial production, the producer price index, and commodity prices; the one-year Treasury rate and the excess bond premium in levels; and the monetary policy shock—and four lags of the firm's own log-capital growth and log-sales growth. The aggregate controls enter through interactions with $S_{j,t-1}$, while the firm-level controls enter both as standalone terms and interacted with $S_{j,t-1}$.

In Panel (b), we additionally control for a richer set of firm characteristics and allow the response to the monetary policy shock to vary with each of them. Specifically, we add one lag of log total assets, Tobin's Q, leverage, age, the quick ratio, dividends, and distance to default, and interact $\hat{\varepsilon}_t$ with each of these lagged characteristics. These interactions account for dimensions of heterogeneity emphasized in the literature, including leverage, default risk, balance-sheet liquidity, firm size, and age (Ottonello and Winberry, 2020; Jeenas, 2024; Gertler and Gilchrist, 1994; Cloyne et al., 2023). Allowing the response to the shock to vary with these characteristics means that δ_h is estimated net of these channels, rather than attributing their effects to the firm's inventory position.

In both specifications the response of tangible capital is negative at every horizon. Low-inventory firms appear to reduce capital stock between 1.5 and 2.5 percent more than their industry peers from roughly the second year onward, with the 90% band excluding zero over most of that range. The responses are similar across the two panels. Thus, the inventory-state amplification is not absorbed by leverage, liquidity, size, or default risk. Because the industry$\times$quarter fixed effects absorb every aggregate and industry-

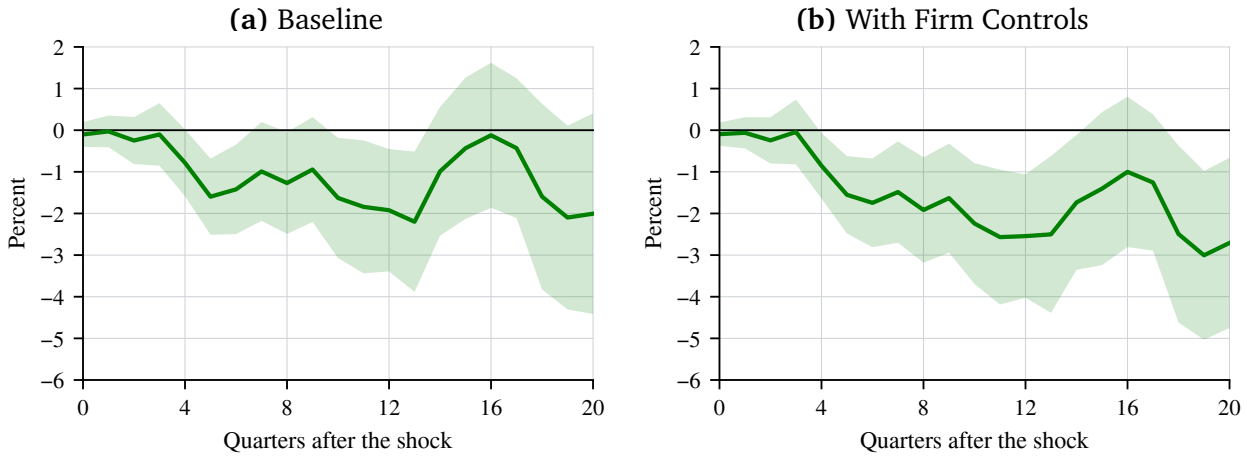


Figure 6: Heterogeneous effects to monetary policy shocks on tangible capital

Note: The coefficient δ_h from the local projection (6): the differential cumulative response of (log) tangible capital of low-inventory firms relative to high-inventory firms in the same industry and quarter. The horizon is in quarters, and the response is the long difference $k_{j,t+h} - k_{j,t-1}$ in percent per 25 basis points. Panel (b) adds lagged firm characteristics and their interactions with the shock, following Ottonello and Winberry (2020) and Jeenas (2024). Both panels include firm and industry×quarter fixed effects, standard errors two-way clustered by firm and quarter, and 90% confidence bands.

wide time-varying force, this evidence points to the firm's own inventory position as the operative margin and corroborates, at the most disaggregated level, the heterogeneity in the responses reported in the aggregate and sectoral analyses.

The heterogeneity is sizeable. In the estimation sample, the average inventory-to-sales ratio is 0.093 for low-inventory firms and 0.207 for high-inventory firms. The difference, 0.114, is about one standard deviation of the ratio (0.109). The estimate therefore describes what a difference of about one standard deviation in the inventory position implies for capital accumulation. For comparison, Jeenas (2024) finds that two years after a one-standard-deviation monetary policy shock, which he equates to a 25 basis point increase in the federal funds rate, a ten percentage point lower liquidity ratio predicts about 0.3 percentage points lower cumulative capital growth. The standard deviation of the liquidity ratio in his sample is 18.5 percentage points, which implies about 0.56 percentage points for a one-standard-deviation difference. At the same two-year horizon, our estimates are

1.3 percent in the baseline specification and 1.9 percent with firm controls, between two and three and a half times as large.

5 Model

We describe a general equilibrium model that generates the state dependence documented above. The model builds on Kryvtsov and Midrigan (2013). The key assumption is that firms choose price and production before their idiosyncratic demand is realized, and hold inventories as a precaution against stocking out. The response of the stockout probability drives the differences in the monetary policy transmission. The stockout probability determines both the value of an additional unit of finished-goods, which affects production decisions, and the price elasticity of demand, which changes the desired markup. In high inventory states, a monetary contraction has a relatively small effect on the stockout probability, provided that the shock is reasonably small, while it substantially reduces the stockout probability in low inventory ones, leading to a greater fall in both production and prices.

5.1 Environment and timing

Time is discrete. There is a representative household, a continuum of monopolistically competitive firms, and a monetary authority. Events unfold as follows. First, aggregate states and the monetary innovation are observed. Firm i inherits depreciated finished-goods inventory $(1-\delta)n_{i,t-1}$ and chooses price P_{it} and labor h_{it}. Production y_{it} becomes available for sale, so total availability is $a_{it} = y_{it} + (1-\delta)n_{i,t-1}$. After choosing production and prices, outlet-level demand shocks are realized, leading to rationing in some outlets and unsold goods in others, which, in turn, become end-of-period inventory n_{it}.

5.2 Demand and sales

Firm i operates a continuum of outlets indexed by j. The firm's notional demand is

$$x_{it} = X_t \left(\frac{P_{it}}{P_t} \right)^{-\theta}, \tag{7}$$

where X_t is aggregate notional demand and $\theta > 1$ is the primitive price elasticity. Notional demand at outlet j is $x_{it}\varepsilon_{ijt}$, where the idiosyncratic demand shock ε_{ijt} is lognormal:

$$\log \varepsilon_{ijt} \sim N\left(-\frac{\sigma_d^2}{2}, \sigma_d^2\right), \qquad \mathrm{E}\varepsilon_{ijt} = 1. \tag{8}$$

Each outlet receives the same availability allocation a_{it}. Actual outlet sales obey the stockout constraint

$$q_{ijt} = \min\{x_{it}\varepsilon_{ijt}, a_{it}\}. \tag{9}$$

In equilibrium, there will be unabsorbed excess demand in some outlets. Due to a continuum of outlets, we can apply the law of large numbers, and firm-level sales simplify to expected outlet sales. Writing $\mathcal{N}(\cdot)$ for the standard normal CDF, expected sales are

$$\begin{aligned} S(a_{it}, x_{it}) &= \mathrm{E}\min\{x_{it}\varepsilon_{ijt}, a_{it}\} \\ &= x_{it}\,\mathrm{E}[\varepsilon_{ijt}\mathbf{1}\{x_{it}\varepsilon_{ijt} < a_{it}\}] + a_{it}\Pr(x_{it}\varepsilon_{ijt} > a_{it}) \\ &= x_{it}\mathcal{N}(d_{1,it}) + a_{it}\left[1 - \mathcal{N}(d_{0,it})\right]. \end{aligned} \tag{10}$$

where

$$d_{0,it} \equiv \frac{\log k_{it} + \sigma_d^2/2}{\sigma_d}, \qquad d_{1,it} \equiv \frac{\log k_{it} - \sigma_d^2/2}{\sigma_d}, \text{ with } k_{it} \equiv \frac{a_{it}}{x_{it}}. \tag{11}$$

The two derivatives used by the model have particularly simple forms:

$$S_a(a, x) = 1 - \mathcal{N}(d_0) = \Pr(x\varepsilon > a), \tag{12}$$

$$S_x(a, x) = \mathcal{N}(d_1). \tag{13}$$

S_a is the stockout probability, and it is also equivalent to the probability that an additional unit of availability is sold. We denote it by $p_{it} \equiv S_a(a_{it}, x_{it})$. Holding availability fixed,

$$\frac{\partial p_{it}}{\partial \log x_{it}} = \frac{\phi_N(d_0)}{\sigma_d} > 0, \tag{14}$$

so a fall in notional demand lowers the probability that the marginal unit is sold. The size of this effect depends on the availability ratio k, which is the source of state dependence described in Section 5.5.

5.3 The firm's problem

Production uses labor according to

$$y_{it} = z_t h_{it}^{1-\alpha}, \qquad 0 \leq \alpha < 1, \tag{15}$$

where z_t is aggregate productivity.

Goods available before current sales and end-of-period inventory are

$$a_{it} = y_{it} + (1-\delta) n_{i,t-1}, \tag{16}$$

$$n_{it} = a_{it} - S(a_{it}, x_{it}). \tag{17}$$

Let

$$g_{it} \equiv \frac{P_{it}}{\bar{\pi} P_{i,t-1}} \tag{18}$$

be the firm's gross price change relative to steady inflation $\bar{\pi}$. We assume firms face the following price adjustment cost

$$\mathcal{C}_{it}^{p} = \frac{\varphi_p}{2}(g_{it} - 1)^2 S_t, \tag{19}$$

where the individual firm takes aggregate sales S_t as given.

Let $r_{it} \equiv P_{it}/P_t$ be the firm's relative price, and collect the aggregate state in Ω_t. The firm's recursive problem can be written as

$$V(n_{i,t-1}, P_{i,t-1}; \Omega_t) = \max_{P_{it}, h_{it}} \left\{ r_{it} S(a_{it}, x_{it}) - \frac{W_t}{P_t} h_{it} - \frac{\varphi_p}{2}(g_{it} - 1)^2 S_t \right.$$

$$+ \mathrm{E}_t[M_{t,t+1} V(n_{it}, P_{it}; \Omega_{t+1})] \Big\}, \tag{20}$$

subject to (7), (15), (16), and (17).

Here W_t is the nominal wage, $P_t = \left(\int_0^1 P_{it}^{1-\theta}\, di\right)^{1/(1-\theta)}$ is the price index, and $\pi_t = P_t/P_{t-1}$ is gross inflation. Aggregate sales, production and hours are $S_t = \int_0^1 S(a_{it}, x_{it})\, di$, $Y_t = \int_0^1 y_{it}\, di$ and $H_t = \int_0^1 h_{it}\, di$. Firms are identical, and in equilibrium $P_{it} = P_t$.

The first order conditions are:

$$mc_{it} = r_{it}\, p_{it} + (1 - p_{it})\, \ell_{it} \tag{21}$$

$$\frac{W_t}{P_t} = (1-\alpha) z_t h_{it}^{-\alpha} mc_{it} \tag{22}$$

$$\begin{aligned} \varphi_p(g_{it} - 1)\frac{g_{it} S_t}{P_{it}} = {} & \frac{S(a_{it}, x_{it})}{P_t} - \frac{\theta x_{it} S_x(a_{it}, x_{it})}{P_{it}}(r_{it} - \ell_{it}) \\ & + \frac{1}{P_{it}} \mathrm{E}_t[M_{t,t+1} \varphi_p(g_{i,t+1} - 1) g_{i,t+1} S_{t+1}]\,. \end{aligned} \tag{23}$$

The term ℓ_{it} is the shadow value of an extra unit of inventory, and it is defined as

$$\ell_{it} \equiv (1-\delta)\mathrm{E}_t[M_{t,t+1} mc_{i,t+1}]\,, \tag{24}$$

where $M_{t,t+1}$ is the household's real stochastic discount factor.

The ability to store finished goods separates the decision of how much to produce from the decision of how much to sell.

Equation (21) is the Euler equation for inventory and determines production. The firm produces until the real marginal cost of production mc_{it} equals the expected value of an additional unit of availability. With probability p_{it} the unit is sold today at the relative price r_{it}; otherwise it is carried into the next period and is worth the shadow value of inventory ℓ_{it}.

Equation (23) is the price-setting condition and determines sales. Each additional sale earns r_{it} but uses a unit that would otherwise be worth ℓ_{it} in inventory, so it adds $r_{it} - \ell_{it}$. Since ℓ_{it} is next period's discounted marginal cost (24), the price depends on expected future costs even when prices are flexible ($\varphi_p = 0$).

5.4 Households, market clearing, and monetary policy

The rest of the model is standard. A representative household chooses consumption, hours and nominal bond holdings to maximize

$$E_0 \sum_{t=0}^{\infty} \beta^t \left[\log C_t - \psi \frac{H_t^{1+\eta}}{1+\eta} \right], \tag{25}$$

where η is the inverse Frisch elasticity. The household supplies labor at the nominal wage W_t, receives firm profits net of price-adjustment costs as nominal dividends, and buys one-period nominal bonds B_t repaid at the gross rate R_t. The period budget constraint is

$$P_t C_t + B_t \;\leq\; R_{t-1} B_{t-1} + W_t H_t + D_t. \tag{26}$$

The first order conditions for hours and bonds are

$$\psi H_t^{\eta} = \frac{1}{C_t} \frac{W_t}{P_t}, \tag{27}$$

$$\frac{1}{C_t} = \beta R_t E_t \left[\frac{1}{C_{t+1}} \frac{P_t}{P_{t+1}} \right]. \tag{28}$$

Equation (27) equates the real wage, valued at the marginal utility of consumption, to the marginal disutility of hours. The real stochastic discount factor is

$$M_{t,t+1} = \beta \frac{C_t}{C_{t+1}}. \tag{29}$$

Bonds are in zero net supply, $B_t = 0$, so (26) reduces to the resource constraint and R_t restricts only the path of C_t.

Monetary policy follows a demeaned Taylor rule:

$$\widehat{R}_t = \rho_R \widehat{R}_{t-1} + (1-\rho_R)\left(\phi_\pi \widehat{\pi}_t + \phi_y \widehat{Y}_t\right) + mp_t, \qquad mp_t = \rho_{mp} mp_{t-1} + \varepsilon_t^R, \tag{30}$$

where Y_t is aggregate production and hats denote log deviations from the deterministic steady state. R_t enters the firm's problem only through $M_{t,t+1}$ in (29), which prices the carried unit in (24), and through aggregate demand X_t.

Actual sales finance consumption and price-adjustment resources:

$$S_t = C_t + \frac{\varphi_p}{2}\left(\frac{\pi_t}{\bar{\pi}} - 1\right)^2 S_t. \tag{31}$$

Aggregate notional demand X_t adjusts so that expected actual sales from (10) satisfy (31). It can be interpreted as purchase attempts or search intensity in a rationed goods market.[11]

5.5 Mechanism

The inventory Euler equation (21) represents the key departure from the textbook Real Business Cycle model. Write $M_{t,t+j} \equiv \prod_{s=0}^{j-1} M_{t+s,t+s+1}$ with $M_{t,t} = 1$. Substituting (24) into (21) and iterating forward,

$$mc_{it} = \mathrm{E}_t \sum_{j \geq 0} (1-\delta)^j M_{t,t+j} \left[\prod_{i'=0}^{j-1} (1 - p_{i,t+i'})\right] p_{i,t+j}\, r_{i,t+j} = \mathrm{E}_t[(1-\delta)^\tau M_{t,t+\tau}\, r_{i,t+\tau}]\,. \tag{32}$$

The marginal cost of production equals the expected discounted price at which the marginal unit is eventually sold. The bracketed product is the probability that the unit is still unsold at $t+j$ and $p_{i,t+j}$ the probability that it sells then, so the weights form the distribution of the waiting time τ until the unit sells, a duration with hazard $p_{i,t+j}$. Proposition 1 char-

[11] The household chooses consumption through (28). X_t is not a separate choice: it is the level of notional demand at which firms' expected sales deliver that consumption. Demand rationed at an outlet is lost; it is not redirected to other outlets or carried into later periods.

acterizes the relation between the path of stockout probabilities and the current marginal cost.

Proposition 1 (Marginal cost and stockout risk) *Hold the path of the stochastic discount factor fixed and suppose* $(1-\delta)M_{t+j,t+j+1} < 1$ *at every date. Then* mc_{it} *is strictly increasing in the path of stockout probabilities: raising* $p_{i,t+j}$ *for some* $j \geq 0$*, weakly for the rest, strictly raises* mc_{it}*.*

The proof is in Appendix F.1.

Consider a monetary tightening. Two channels lower the marginal cost of production. The first is contemporaneous. By (14), lower notional demand reduces the probability that the marginal unit of availability sells today, and by Proposition 1 a lower selling probability lowers the value of that unit and hence marginal cost. The second is intertemporal. A higher real rate lowers the discount factors in (32), so the future sales of a unit that stays on the shelf are worth less today. Through (22), lower marginal cost shifts labor demand down at a given wage. Lower consumption raises labor supply, so hours fall if the shift in labor demand dominates. This is how inventories address the comovement problem of Barro and King (1984). Without inventories and with flexible prices, marginal cost is pinned down by the markup, a demand disturbance cannot shift labor demand, and consumption and hours move in opposite directions. With inventories, the marginal cost of production is the value of a unit on the shelf, and that value depends on demand.

Turning to the price dynamics, the price-setting condition under flexible prices becomes

$$r_{it} = \frac{\epsilon_{it}^{\text{eff}}}{\epsilon_{it}^{\text{eff}} - 1}\ell_{it}, \tag{33}$$

where $\epsilon_{it}^{\text{eff}}$ is the effective price elasticity and it is defined as

$$\epsilon_{it}^{\text{eff}} \equiv -\frac{\partial \log S(a_{it}, x_{it})}{\partial \log P_{it}} = \theta \frac{x_{it} S_x(a_{it}, x_{it})}{S(a_{it}, x_{it})}. \tag{34}$$

Unlike in the textbook New Keynesian model, the firm marks up over the shadow value of inventory ℓ_{it}, the value of the unit it gives up by selling today, rather than over the marginal cost of production. The markup also differs from the CES benchmark $\theta/(\theta-1)$. At outlets that stock out, sales equal availability and do not respond to price, so a price cut raises sales only at the remaining outlets. The relevant elasticity is therefore the effective price elasticity $\epsilon_{it}^{\text{eff}}$ in (34), which is below θ whenever stockouts occur. Proposition 2 shows that it depends on the firm's state only through the stockout probability and falls as that probability rises.

Proposition 2 (Effective elasticity and stockout risk) *The effective price elasticity depends on (a_{it}, x_{it}) only through the stockout probability p_{it}:*

$$\frac{\epsilon_{it}^{\text{eff}}}{\theta} = \frac{\mathcal{N}(d_{1,it})}{\mathcal{N}(d_{1,it}) + k_{it} p_{it}} = \frac{\mathrm{E}[x_{it}\varepsilon_{ijt}\,\mathbf{1}\{x_{it}\varepsilon_{ijt} < a_{it}\}]}{S(a_{it}, x_{it})}, \tag{35}$$

where $d_{0,it} = \mathcal{N}^{-1}(1-p_{it})$, $d_{1,it} = d_{0,it} - \sigma_d$ and $k_{it} = e^{\sigma_d d_{0,it} - \sigma_d^2/2}$. Thus $\epsilon_{it}^{\text{eff}}/\theta$ is the share of expected sales made at outlets that do not stock out. For every $\sigma_d > 0$, the map $p \mapsto \epsilon^{\text{eff}}$ is strictly decreasing on $(0,1)$, with $\epsilon^{\text{eff}} \to \theta$ as $p \to 0$ and $\epsilon^{\text{eff}} \to 0$ as $p \to 1$.

The proof is in Appendix F.1.

With price adjustment costs, log-linearizing (23) around the zero-inflation symmetric steady state gives the Phillips curve

$$\widehat{\pi}_t = \beta \mathrm{E}_t \widehat{\pi}_{t+1} + \frac{\epsilon^{\text{eff},*} - 1}{\varphi_p}\left(\widehat{\mu}_t + \widehat{\ell}_t\right), \qquad \mu_t \equiv \frac{\epsilon_t^{\text{eff}}}{\epsilon_t^{\text{eff}} - 1}. \tag{36}$$

Inflation is driven by the desired price $\mu_t \ell_t$: a markup over the shadow value of inventory. A monetary tightening that lowers stockout risk lowers both terms. Fewer outlets are rationed, the effective price elasticity rises and the desired markup falls; lower future stockout probabilities and a higher real rate reduce the continuation value.

5.6 State dependence

The model generates the state dependence documented in Sections 2–4. By (32) and (36), the production and price responses to a monetary shock depend on two paths: the real interest rate, through the stochastic discount factor, and the stockout probability. The real-rate path differs across the low- and high-inventory states only through the endogenous feedback of policy to inflation and output. The stockout path, in contrast, is directly affected by the inventory position: by (14), the response of the stockout probability to notional demand vanishes as $k \to \infty$, and peaks when the stockout probability is one half ($p = 1/2$). On the range $p < 1/2$, which contains our calibration, a lower availability ratio implies a larger response of the stockout probability to a given fall in notional demand.

The limit $k \to \infty$ shows what the stockout margin adds. No outlet is rationed: $p \to 0$, $S \to x$, and a change in notional demand passes one for one into sales. The stockout probability no longer responds to demand, and by Proposition 2 the effective elasticity equals the CES benchmark θ. Two channels become absent. Marginal cost loses its hazard channel and moves only with the discount path. The markup term in (36) drops out, so inflation moves only with ℓ_t, which by (21) equals mc_{it} when $p_{it} = 0$. In contrast, these two channels operate in the low-inventory state, so a tightening lowers the stockout probability, marginal cost, labor demand and inflation by more when inventories are low.

5.7 Calibration

The model period is one month. Table 1 lists the parameters set directly, and Table 2 the parameters that discipline inventories and stockouts.

The discount factor implies an annual real interest rate of four percent. The Taylor rule responds to inflation with a coefficient of 1.5 and does not respond to output. The Rotemberg coefficient implies the same slope of the Phillips curve as a Calvo model in which a price survives a month with probability 0.93, an expected price duration of about 14

months.[12] The inverse Frisch elasticity of labor supply is 2, and the curvature of production, $\alpha = 0.2$, makes the marginal cost of production increasing in output. The policy rate is smoothed with a monthly coefficient of 0.9. The persistence of the monetary disturbance, $\rho_{mp} = 0.88$, is set so that the policy rate returns to zero about a year after the shock.

Table 1: Parameters set directly (monthly)

Parameter		Value
A. Conventional values		
β	discount factor	0.9967
ϕ_π	inflation coefficient	1.50
ϕ_y	output coefficient	0
$\bar{\pi}$	steady gross inflation	1
φ_p	Rotemberg coefficient	630
B. Other parameters		
η	inverse Frisch elasticity	2.00
α	production curvature	0.20
ρ_R	rate smoothing	0.90
ρ_{mp}	persistence of the policy disturbance	0.88

The inventory block is disciplined by three values reported in Kryvtsov and Midrigan (2013): a monthly depreciation rate of inventories of 1.1 percent, about 12 percent a year, a stockout frequency of 5 percent, and an inventory-sales ratio of 1.4 months of sales. We set $\delta = 0.011$ directly. The primitive elasticity θ and the dispersion of outlet demand σ_d are chosen jointly so that the steady-state stockout probability is $p^* = 0.05$ and the inventory-sales ratio is 1.4. The availability ratio is then

$$k^* = \exp\left(\sigma_d \mathcal{N}^{-1}(1 - p^*) - \tfrac{1}{2}\sigma_d^2\right) = 2.306. \tag{37}$$

[12] This mapping treats the firm's marginal cost as independent of its own output. With decreasing returns ($\alpha > 0$), the Calvo slope is scaled by $(1-\alpha)/(1-\alpha+\alpha\epsilon^{\text{eff},*})$, and the same slope corresponds to a monthly survival probability of 0.90, an expected duration of about 10 months. The inventory buffer weakens the link between a firm's sales and its production, so the relevant duration lies between the two. Micro estimates for regular prices imply durations of about 8 to 11 months (Nakamura and Steinsson, 2008).

Table 2: Inventory block: targets from Kryvtsov and Midrigan (2013)

Target	Value	Parameter	Value
Inventory depreciation (monthly)	0.011	δ, inventory depreciation	0.011
Stockout frequency	0.05	θ, primitive elasticity	5.05
Inventory-sales ratio (months)	1.40	σ_d, dispersion of outlet demand	0.628

δ is set directly. θ and σ_d are chosen jointly to match the stockout frequency and the inventory-sales ratio.

Table 3 reports the implied steady state. With zero inflation the pricing condition (33) holds with $r^* = 1$, so the effective elasticity is $\epsilon^{\text{eff},*} = 1/(1-\ell^*)$. Since ℓ^* depends only on p^*, β and δ, so do the effective elasticity, $\epsilon^{\text{eff},*} = 4.44$, and the markup over the shadow value of inventory, 29 percent. At the calibrated state, 12 percent of expected sales are made at outlets that stock out. By Proposition 2, the primitive elasticity is the effective elasticity divided by the share of sales made at the remaining outlets, $\theta = 4.44/0.88 = 5.05$.

Table 3: Implied steady state

Object	Value	Object	Value
k^*, availability ratio	2.306	$\epsilon^{\text{eff},*}$, effective elasticity	4.44
f^*, fill rate	0.961	$\epsilon^{\text{eff},*}/(\epsilon^{\text{eff},*}-1)$, markup	1.29
mc^*, marginal cost	0.786	y^*/S^*, production-sales ratio	1.015
ℓ^*, shadow value of inventory	0.775	h^*, hours	0.861

5.8 Impulse responses by initial inventory position

We solve the model at third-order and compute conditional generalized impulse responses by simulating the dynamics starting from two initial conditions—one with a high inventory-to-sales ratio and one with a low ratio. The initial finished-goods inventory-sales states are defined as

$$(I/S)_L = (I/S)^* e^{-g}, \qquad (I/S)_H = (I/S)^* e^{g}, \tag{38}$$

with $g = 0.40$ in the baseline. We measure the difference in variable paths with and without a contractionary monetary policy shock. The shock is normalized so that the policy rate rises by 25 basis points on impact.

Table 4 reports where the two states sit on the stockout schedule. A log gap of 0.40 in the inventory-sales ratio maps into a stockout probability of 0.098 in the low state and 0.021 in the high state.

Table 4: Conditional states at $g = 0.40$

	Low I/S	Steady state	High I/S
Inventory-sales ratio, I/S (months)	0.938	1.400	2.089
Availability ratio, k	1.854	2.306	2.966
Stockout probability, p	0.098	0.050	0.021
Effective elasticity, ϵ^{eff}	4.07	4.44	4.74

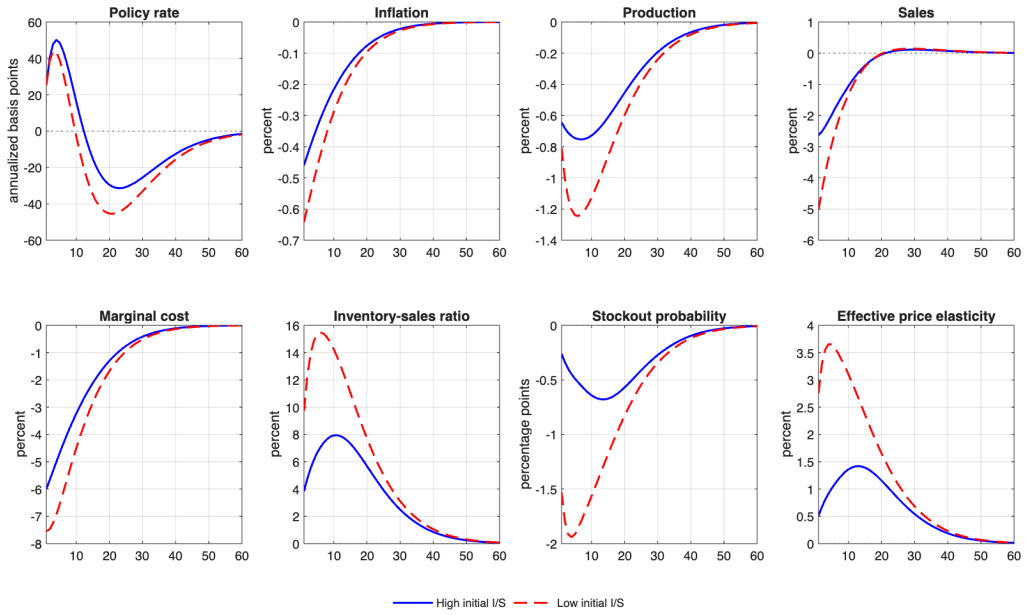


Figure 7: Conditional generalized impulse responses to a contractionary monetary shock, normalized to a 25 basis point increase in the policy rate on impact in each state. Horizontal axis in months. Solid: high initial inventory-sales ratio. Dashed: low initial ratio.

Figure 7 plots the conditional responses. A monetary tightening lowers production, sales and inflation in both states, and by more when inventories are low. In the low state the stockout probability falls and the effective elasticity rises; in the high state both change much less.

Table 5 measures the impact contributions of the channels discussed in Section 5.5. Specifically we isolate the contributions of the path of the stochastic discount factor and the stockout probability to the change in the marginal cost, as per equation (32). We find that while the discount path governs most of the response in the high state, the difference between states is explained by the responses of the stockout probability. Furthermore, we perform a similar exercise analyzing the path of the desired price response. As shown by the Phillips Curve in equation (36), this is simply the sum of the markup and the shadow value of inventory, which is affected by the future marginal cost. Table 5 shows that the future path in the marginal cost drives the difference in the response, while the markup only contributes to a quarter.

Table 5: Impact responses by channel (percent)

	Low I/S	High I/S	Low − High
Marginal cost, mc_t			
Stockout path $\{p\}$	-4.44	-2.99	-1.45 (118%)
Discount path $\{M\}$	-3.02	-3.40	$+0.38$ (-31%)
Interaction	-0.17	-0.01	-0.16 (13%)
Total	-7.63	-6.40	-1.23
Desired price, $\mu_t \ell_t$			
Markup, μ_t	-0.87	-0.14	-0.73 (26%)
Marginal cost, ℓ_t	-8.39	-6.29	-2.10 (74%)
Total	-9.26	-6.42	-2.83

Shares of the difference between states in parentheses.

6 Conclusion

This paper shows that the inventory position of firms shapes the transmission of monetary policy. A contractionary monetary policy shock lowers industrial production and producer prices more than twice as much when the economy enters the episode with a low inventory-to-sales ratio as when it enters with a high one; in the high-inventory regime the responses are small and statistically insignificant. We document this asymmetry at three levels of aggregation. In an aggregate threshold VAR identified with external instruments,

it survives a wide range of choices for the instrument, filter, sample, and variable specification. In a panel of manufacturing industries, it survives time fixed effects, which remove the business cycle and every other force common to a month. In firm-level data, firms whose inventories are low relative to their industry peers cut their capital stock by more than comparable firms in the same industry and quarter, and the difference is not absorbed by leverage, liquidity, size, or default risk.

These findings qualify a long-standing conclusion. Because production-smoothing models fail to match the volatility of production and the countercyclical inventory-to-sales ratio, the literature has largely treated inventories as marginal to aggregate fluctuations. Our results show that, whatever their role in unconditional fluctuations, inventories shape the economy's response to monetary policy. To rationalize the asymmetry, we embed the stockout-avoidance motive of Kryvtsov and Midrigan (2013) in an otherwise standard New Keynesian model. The probability of stocking out determines both the value of an additional unit of goods and the price elasticity of demand. A monetary tightening lowers that probability, which lowers the value of the marginal unit, labor demand, and the price firms want to set. When inventories are ample, stockouts are rare and a fall in demand barely changes their probability, so production and prices respond less.

The results carry implications for the conduct of monetary policy. The strength of the transmission mechanism varies with an observable state variable, the inventory-to-sales ratio: a given tightening is considerably more contractionary, for both output and prices, when inventories are lean than when they are abundant. The inventory position thus joins capacity utilization and supply-chain conditions among the real-side state variables that govern monetary transmission. Unlike those, which mainly change how a policy action divides between output and prices, the inventory position changes the size of both responses.

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A Data

A.1 Aggregate data

The table below lists the aggregate variables used throughout the paper.

One-year Treasury bill rate. *Source:* FRED-MD Monthly Macroeconomic Database; *Code:* GS1; *Reference:* McCracken and Ng (2016); *Transformation:* Percent per annum; normalized to basis points in figures.

Industrial production. *Source:* FRED-MD Monthly Macroeconomic Database; *Code:* INDPRO; *Reference:* McCracken and Ng (2016); *Transformation:* Logarithmic.

Producer price index. *Source:* FRED-MD Monthly Macroeconomic Database; *Code:* PPIFGS (Producer price index: finished goods); *Reference:* McCracken and Ng (2016); *Transformation:* Logarithmic.

Commodity prices. *Source:* FRED-MD Monthly Macroeconomic Database; *Code:* CUSR0000SAC (CPI for all urban consumers: commodities); *Reference:* McCracken and Ng (2016); *Transformation:* Logarithmic.

Excess bond premium. *Source:* Federal Reserve Board website; *Code:* EBP; *Reference:* Gilchrist and Zakrajšek (2012) and extended by Favara et al. (2016); *Transformation:* Percent; normalized to basis points in figures.

Finished-goods inventory-to-sales ratio. *Notes:* Ratio of two objects. *Transformation:* Band-pass filtered cyclical component (periodicities 2–120 months). **Numerator** Finished-goods inventories is the sum of finished-goods inventories of the manufacturing sector, total inventories of the wholesale sector, and total inventories of the retail sector, drawn respectively from the Census Bureau's M3 (U.S. Census Bureau, 2023a), MWTS (U.S. Census Bureau, 2023c), and MRTS (U.S. Census Bureau, 2023b) surveys (all seasonally adjusted).[13] **Denominator** Current-quarter total sales are defined as total business sales,

[13] The Census Bureau redesigned these surveys from the Standard Industrial Classification (SIC) to the North American Industry Classification System (NAICS) in the early 2000s, so NAICS-based data are available only

recovered from McCracken and Ng (2016) as $\texttt{BUSINVx}_t/\texttt{ISRATIOx}_t$, computed as the three-month cumulative sum $s_{t-2} + s_{t-1} + s_t$.

Consumer price index. *Source:* FRED-MD Monthly Macroeconomic Database; *Code:* CPIAUCSL (CPI for all urban consumers: all items); *Reference:* McCracken and Ng (2016); *Transformation:* Logarithmic.

Backlogs of orders. *Source:* US Census Bureau, Monthly Survey of Manufacturers (M3); *Reference:* U.S. Census Bureau (2023a); *Coverage:* Manufacturing sector only; *Transformation:* In the robustness check, backlogs of orders are subtracted from finished-goods manufacturing inventories in the numerator of the inventory-to-sales ratio as described above; the denominator remains unchanged.

Unemployment rate. *Source:* FRED-MD Monthly Macroeconomic Database; *Code:* UNRATE (civilian unemployment rate); *Reference:* McCracken and Ng (2016); *Transformation:* Percentage points.

A.2 Sector-level data

The sectoral dataset is a monthly panel of the 21 three-digit NAICS manufacturing sectors (NAICS 311–339), observed from January 1992 to December 2023. We restrict attention to manufacturing because it is the only part of the economy for which all three series are published at the three-digit level: the M3 survey reports inventories by stage of fabrication, and industrial production is not defined for the wholesale and retail sectors. The panel is unbalanced because PPI coverage begins after January 1992 for many sectors. All quantity and inventory series are seasonally adjusted. PPI series are not seasonally adjusted, as the BLS does not publish seasonally adjusted three-digit NAICS producer price indices.

Manufacturers' Shipments, Inventories, and Orders (M3). *Source:* U.S. Census Bureau; *Reference:* U.S. Census Bureau (2023a); *Coverage:* 21 manufacturing sectors (NAICS 311–339), January 1992 – December 2023. The M3 survey provides, for each sector:

from that point onward. In order to have data since December 1980, we extend the series by matching the corresponding historical SIC-based data to the NAICS-based series.

value of shipments (SALES), new orders (NewOrders), unfilled orders (UnfilledOrders), total inventories (TotInv), finished-goods inventories (OutInv), materials and supplies inventories (MatInv), work-in-process inventories (WIPInv), and the inventory-to-sales ratio (ISRatio). All series are seasonally adjusted.

Industrial Production and Capacity Utilization (G.17). *Source:* Board of Governors of the Federal Reserve System; *Reference:* Board of Governors of the Federal Reserve System (2023); *Coverage:* 21 manufacturing sectors (NAICS 311–339), January 1992 – ongoing. The G.17 release provides monthly industrial production indices (IP) and capacity utilization indices (CAP) for each manufacturing sector. These series are not available for wholesale or retail sectors.

Producer price index, sector data. *Source:* U.S. Bureau of Labor Statistics; *Reference:* U.S. Bureau of Labor Statistics (2023); *Coverage:* 21 manufacturing sectors. Series are not seasonally adjusted. The BLS publishes these three-digit NAICS aggregate series from different start dates: for 9 manufacturing sectors (NAICS 311, 316, 324, 325, 326, 327, 331, 332, 337), coverage begins in January 1992; for the remaining 12 manufacturing sectors, coverage begins in December 2003; for the 2 wholesale sectors (NAICS 423 and 424), coverage begins in June 2004 and June 2005, respectively; for NAICS 441, coverage begins in December 2003. Retail sectors NAICS 442–454 have no published three-digit series; PPI is set to NaN for these sectors throughout. To ensure comparability across months within each sector, PPI is sourced exclusively from the BLS three-digit aggregate series; no mixing of methods is applied.

Table 6 lists all 21 sectors, their data sources by variable group, and the PPI coverage start date.

Table 6: Sectoral panel dataset: sector coverage and variable sources

NAICS	Description	IP	Inv./Sales	PPI	PPI start
Panel A: Manufacturing (21 sectors) — M3 + G.17, January 1992 – December 2023					
311	Food Manufacturing	G.17	M3	BLS	Jan 1992
312	Beverage and Tobacco Product Mfg	G.17	M3	BLS	Dec 2003
313	Textile Mills	G.17	M3	BLS	Dec 2003
314	Textile Product Mills	G.17	M3	BLS	Dec 2003
315	Apparel Manufacturing	G.17	M3	BLS	Dec 2003
316	Leather and Allied Product Mfg	G.17	M3	BLS	Jan 1992
321	Wood Product Manufacturing	G.17	M3	BLS	Dec 2003
322	Paper Manufacturing	G.17	M3	BLS	Dec 2003
323	Printing and Related Support Activities	G.17	M3	BLS	Dec 2003
324	Petroleum and Coal Products Mfg	G.17	M3	BLS	Jan 1992
325	Chemical Manufacturing	G.17	M3	BLS	Jan 1992
326	Plastics and Rubber Products Mfg	G.17	M3	BLS	Jan 1992
327	Nonmetallic Mineral Product Mfg	G.17	M3	BLS	Jan 1992
331	Primary Metal Manufacturing	G.17	M3	BLS	Jan 1992
332	Fabricated Metal Product Mfg	G.17	M3	BLS	Jan 1992
333	Machinery Manufacturing	G.17	M3	BLS	Dec 2003
334	Computer and Electronic Product Mfg	G.17	M3	BLS	Dec 2003
335	Electrical Equipment, Appliance & Component	G.17	M3	BLS	Dec 2003
336	Transportation Equipment Mfg	G.17	M3	BLS	Dec 2003
337	Furniture and Related Product Mfg	G.17	M3	BLS	Jan 1992
339	Miscellaneous Manufacturing	G.17	M3	BLS	Dec 2003

Notes: IP and CAP (G.17) are available for manufacturing only. PPI "—" denotes no published BLS three-digit NAICS series.

Estimation sample. The local projection of Section 3 uses the 21 manufacturing sectors (NAICS 311–339). The wholesale and retail sectors are excluded because the G.17 does not publish industrial production for sectors that distribute rather than produce goods, so the left-hand side of equation (4) is undefined for them. Within manufacturing, the panel is unbalanced in the time dimension only, because producer price coverage begins at two different dates: nine sectors (NAICS 311, 316, 324, 325, 326, 327, 331, 332, 337) are observed from January 1992 and the remaining twelve from December 2003. We do not balance the panel, which would discard the first twelve years for the nine early sectors; we estimate on the unbalanced sample and absorb the resulting composition differences with the sector and time fixed effects. Each projection is run on the months in which the sector's IP index and PPI are jointly observed, so that all four outcomes are estimated on

the same observations and their responses are directly comparable. Shipments and the IS ratio impose no further restriction, because the inventory state is itself constructed from them. The impact regression uses $5{,}853$ sector-month observations.

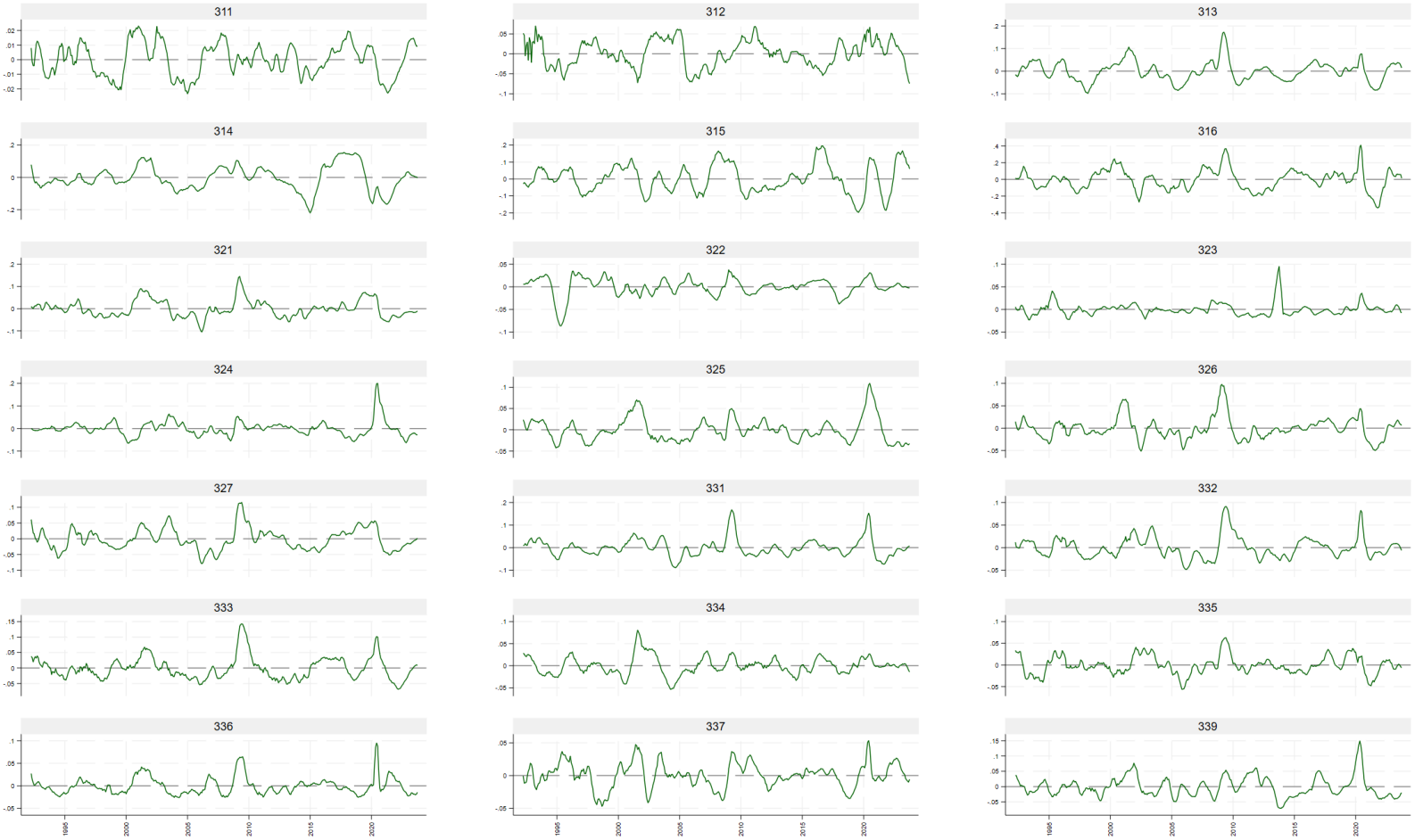


Figure 8: Sector-Level Filtered IS Ratios

Note: This figure plots the sector-level filtered IS ratios with the same high-pass filter as in the aggregate analysis to remove each sector's long-term trend beyond the periodicity of 10 years.

B Threshold VAR: estimation and impulse responses

B.1 Model and posterior sampler

Model. Consider the following threshold VAR model:

$$Y_t = \left[c_1 + \sum_{j=1}^{P} B_{1,j} Y_{t-j} + u_{1,t} \right] S_t + \left[c_2 + \sum_{j=1}^{P} B_{2,j} Y_{t-j} + u_{2,t} \right] (1 - S_t) \tag{B.1}$$

where $Y_t = [y_{1,t},\, y_{2,t},\, \cdots,\, y_{n,t}]'$ is a $n \times 1$ vector of endogenous variables; $u_{i,t} \sim \mathcal{N}(0, \Sigma_i)$ is a $n \times 1$ vector of reduced-form innovations in regime $i \in \{1, 2\}$, with Σ_i denoting the respective $n \times n$ covariance matrix; and c_i and $B_{i,1}, \ldots, B_{i,p}$ are the $n \times 1$ intercept vector and the $n \times n$ autoregressive slope matrices in regime $i = 1, 2$. The regime indicator S_t is defined as:

$$S_t = 1 \iff y_{n,t-d} \leq y_n^*, \tag{B.2}$$

where $y_{n,t}$ is the threshold variable, assumed without loss of generality to be the last element of Y_t, y_n^* is the threshold value, and $d \geq 1$ is the delay parameter governing the minimum lag with which $y_{n,t}$ can affect regime classification. Accordingly, $S_t = 1$ identifies the "low" regime, prevailing when the threshold variable lies below y_n^*, while $S_t = 0$ identifies the "high" regime. The objects of inference are $\boldsymbol{\theta} \equiv \{B_{i,j},\, c_i,\, \Sigma_i\}_{i=1,2;\; j=1,\ldots,p}$ together with the threshold value y_n^* and the delay parameter d.

Notation. Define the $(np + 1)$-dimensional regressor vector $X_t \equiv [Y_{t-1}', \ldots, Y_{t-p}', 1]'$ and the stacked coefficient matrix $\mathbf{B}_i \equiv [B_{i,1}, B_{i,2}, \ldots, B_{i,p}, c_i]' \in \mathbb{R}^{(np+1)\times n}$, so that the VAR in regime i reads $Y_t = \mathbf{B}_i' X_t + u_{i,t}$. Let $\mathcal{T}_1 \equiv \{t : S_t = 1\}$ and $\mathcal{T}_2 \equiv \{t : S_t = 0\}$ be the regime-specific observation sets, with size T_1 and T_2, respectively. Stacking Y_t' and X_t' over $t \in \mathcal{T}_i$ into matrices $\mathbf{Y}_i \in \mathbb{R}^{T_i \times n}$ and $\mathbf{X}_i \in \mathbb{R}^{T_i \times (np+1)}$ yields the system

$$\mathbf{Y}_i = \mathbf{X}_i \mathbf{B}_i + \mathbf{U}_i, \qquad i = 1, 2. \tag{B.3}$$

Prior. Estimation proceeds by Bayesian methods. We adopt a Minnesota prior, implemented via the dummy-observation device of Bańbura et al. (2010). For each variable k, an AR(1) model is first estimated on the data, yielding an autoregressive coefficient δ_k (truncated at one in absolute value) and a residual standard deviation σ_k. The prior sets the mean of each variable's own first-lag coefficient to δ_k and the mean of all remaining coefficients to zero. Prior variances decrease with the lag order, with overall shrinkage governed by a scalar hyperparameter $\lambda > 0$, and are normalized by the ratio σ_j/σ_k to make the prior scale-invariant across variables. Operationally, this component is implemented by appending the following dummy observations to the data of each regime, where $J_p \equiv \mathrm{diag}(1, 2, \ldots, p)$ and $\epsilon > 0$ is a small positive constant that places a nearly flat prior on the intercepts c_i:

$$\mathbf{Y}_{D,1} = \begin{bmatrix} \frac{1}{\lambda}\,\mathrm{diag}(\delta_1\sigma_1,\, \ldots,\, \delta_n\sigma_n) \\ \mathbf{0}_{n(p-1)\times n} \\ \mathrm{diag}(\sigma_1,\, \ldots,\, \sigma_n) \\ \mathbf{0}_{1\times n} \end{bmatrix}, \qquad \mathbf{X}_{D,1} = \begin{bmatrix} \frac{1}{\lambda}\, J_p \otimes \mathrm{diag}(\sigma_1,\, \ldots,\, \sigma_n) & \mathbf{0}_{np\times 1} \\ \mathbf{0}_{n\times np} & \mathbf{0}_{n\times 1} \\ \mathbf{0}_{1\times np} & \epsilon \end{bmatrix}. \tag{B.4}$$

A sum-of-coefficients component with hyperparameter $\tau > 0$ additionally shrinks the sum of each variable's lag coefficients toward one, capturing unit-root behaviour. It is implemented via a second set of dummy observations, where $\mathbf{1}_p$ the p-vector of ones:

$$\mathbf{Y}_{D,2} = \frac{1}{\tau}\,\mathrm{diag}(\delta_1\mu_1,\, \ldots,\, \delta_n\mu_n), \qquad \mathbf{X}_{D,2} = \begin{bmatrix} \mathbf{1}_p' \otimes \frac{1}{\tau}\,\mathrm{diag}(\delta_1\mu_1,\, \ldots,\, \delta_n\mu_n) & \mathbf{0}_{n\times 1} \end{bmatrix}. \tag{B.5}$$

Letting $\widetilde{T} \equiv n(p+2)+1$ denote the total number of dummy observation rows, the augmented data matrices for regime i are:

$$\mathbf{Y}_i^* \equiv \begin{bmatrix} \mathbf{Y}_i \\ \mathbf{Y}_{D,1} \\ \mathbf{Y}_{D,2} \end{bmatrix} \in \mathbb{R}^{T_i^*\times n}, \qquad \mathbf{X}_i^* \equiv \begin{bmatrix} \mathbf{X}_i \\ \mathbf{X}_{D,1} \\ \mathbf{X}_{D,2} \end{bmatrix} \in \mathbb{R}^{T_i^*\times(np+1)}, \tag{B.6}$$

where $T_i^* \equiv T_i + \widetilde{T}$. Importantly, the priors imply a conservative approach: the same dummy observations are appended to both regimes, so that $\mathbf{B}_1$ and $\mathbf{B}_2$ share a common prior that shrinks them toward being identical.

Finally, the threshold value y_n^* is assigned an independent Gaussian prior $\pi(y_n^*) = \mathcal{N}(\bar{y}_n^*, \sigma_n^2)$, with mean $\bar{y}_n^*$ equal to the sample median of the threshold variable, $y_{n,t-d}$, and variance σ_n^2 chosen large enough such that the prior is uninformative. The delay parameter d is assigned a discrete uniform prior over a finite candidate set $\mathcal{D} \subset \mathbb{Z}_+$.

Posterior and Gibbs sampler. The joint posterior $p(\boldsymbol{\theta},\, y_n^*,\, d \mid \mathbf{Y})$ does not admit a closed form. We draw from it via a Markov chain Monte Carlo (MCMC) algorithm that combines Gibbs sampling for the VAR parameters with a Metropolis-Hastings (MH) step for the threshold and a discrete Gibbs step for the delay. Each iteration cycles through the following four steps, conditioning on all remaining parameters.

Step 1 — VAR coefficients. Conditional on Σ_i, y_n^*, and d, the posterior of $\mathbf{B}_i$ follows from the Normal-Inverse-Wishart conjugacy of the augmented system. Specifically:

$$\operatorname{vec}(\mathbf{B}_i) \mid \Sigma_i,\, y_n^*,\, d,\, \mathbf{Y} \;\sim\; \mathcal{N}\Big(\hat{b}_i,\; \Sigma_i \otimes \big(\mathbf{X}_i^{*\prime}\mathbf{X}_i^*\big)^{-1}\Big)\,, \tag{B.7}$$

where $\hat{b}_i \equiv \operatorname{vec}\Big(\big(\mathbf{X}_i^{*\prime}\mathbf{X}_i^*\big)^{-1}\mathbf{X}_i^{*\prime}\mathbf{Y}_i^*\Big)$ is the posterior mean, equal to the ordinary least-squares estimator on the augmented system. Draws from (B.7) are subject to a covariance-stationarity constraint: a candidate draw $\mathbf{B}_i^{(m)}$ is accepted only if all eigenvalues of the associated companion-form matrix lie strictly inside the unit circle; otherwise it is discarded and a new draw is taken (rejection sampling). If no stationary draw is found within a fixed maximum number of attempts, $\mathbf{B}_i$ is held at the most recently accepted value.

Step 2 — Covariance matrices. Conditional on $\mathbf{B}_i$, y_n^*, and d, the posterior of Σ_i is Inverse-Wishart:

$$\Sigma_i \mid \mathbf{B}_i,\, y_n^*,\, d,\, \mathbf{Y} \;\sim\; \mathcal{IW}\Big(\hat{S}_i,\; T_i^*\Big)\,, \tag{B.8}$$

where $\hat{S}_i \equiv \hat{\mathbf{E}}_i^{\prime}\hat{\mathbf{E}}_i$ and $\hat{\mathbf{E}}_i \equiv \mathbf{Y}_i^* - \mathbf{X}_i^*\mathbf{B}_i$ is the matrix of residuals on the augmented data.

Step 3 — Threshold value. Since y_n^* determines the regime partition $(\mathcal{T}_1, \mathcal{T}_2)$, its conditional posterior is non-standard. We employ an MH step with a symmetric Gaussian random-walk proposal: $y_n^{*\,\text{prop}} = y_n^* + \varepsilon$, $\varepsilon \sim \mathcal{N}(0, \sigma_{\text{rw}}^2)$. Because the proposal is symmetric, the Hastings correction cancels and the log-acceptance ratio reduces to:

$$\log r = \log p(y_n^{*\,\text{prop}} \mid \boldsymbol{\theta},\, d,\, \mathbf{Y}) - \log p(y_n^* \mid \boldsymbol{\theta},\, d,\, \mathbf{Y})\,, \tag{B.9}$$

where, up to a normalizing constant,

$$\log p(y_n^* \mid \boldsymbol{\theta}, d, \mathbf{Y}) = \sum_{i=1}^{2} \left[\frac{T_i}{2} \log \left|\Sigma_i^{-1}\right| - \frac{1}{2} \sum_{t \in \mathcal{T}_i} \left(Y_t - \mathbf{B}_i' X_t\right)' \Sigma_i^{-1} \left(Y_t - \mathbf{B}_i' X_t\right) \right] + \log \pi(y_n^*). \tag{B.10}$$

A proposal is immediately rejected if either regime would contain fewer than $\underline{T}$ observations, where $\underline{T}$ is a lower bound imposed to ensure invertibility of the regime-specific regressor matrices $\mathbf{X}_i^{*\prime}\mathbf{X}_i^*$. Otherwise, $y_n^{*\,\text{prop}}$ is accepted with probability $\min(1,\, e^{\log r})$. That is, draw $u \sim \mathcal{U}(0,1)$ and replace y_n^* with $y_n^{*\,\text{prop}}$ if $\log r > \log u$, retaining the current value otherwise.

Step 4 — Delay parameter. The delay d takes values in the finite set $\mathcal{D}$. Its conditional posterior is:

$$p(d = d' \mid \boldsymbol{\theta},\, y_n^*,\, \mathbf{Y}) \;\propto\; p\big(\mathbf{Y} \mid \boldsymbol{\theta},\, y_n^*,\, d'\big) \cdot \pi(d'), \tag{B.11}$$

where $\pi(d')$ is uniform over $\mathcal{D}$. Crucially, when evaluating the likelihood at lag d', the regime partition itself changes: define $\mathcal{T}_1(d') \equiv \{t : y_{n,t-d'} \le y_n^*\}$ and $\mathcal{T}_2(d') \equiv \{t : y_{n,t-d'} > y_n^*\}$, with size $T_1(d')$ and $T_2(d')$. The log-likelihood at lag d' is then:

$$\log p\big(\mathbf{Y} \mid \boldsymbol{\theta},\, y_n^*,\, d'\big) = \sum_{i=1}^{2} \left[\frac{T_i(d')}{2} \log \left|\Sigma_i^{-1}\right| - \frac{1}{2} \sum_{t \in \mathcal{T}_i(d')} \left(Y_t - \mathbf{B}_i' X_t\right)' \Sigma_i^{-1} \left(Y_t - \mathbf{B}_i' X_t\right) \right]. \tag{B.12}$$

The log-weights $\{\log p(\mathbf{Y} \mid \boldsymbol{\theta}, y_n^*, d')\}_{d' \in \mathcal{D}}$ are converted into a discrete probability mass function over $\mathcal{D}$ using the softmax transformation, and a new value of d is drawn from this distribution, so that each candidate delay d' is selected with probability proportional to its posterior likelihood.

Steps 1–4 constitute one full MCMC iteration. The sampler is initialized from the full-sample OLS estimate (on the augmented data) for both coefficient matrices, from the identity matrix for

both covariance matrices, and from the prior mean $\bar{y}_n^*$ for the threshold. After a burn-in of M_{burn} iterations, every k-th draw is retained out of M_{draws} total iterations, yielding a thinned posterior sample of size $(M_{\text{draws}} - M_{\text{burn}})/k$.

B.2 Generalized impulse responses

Our object of interest is the state-dependent dynamic response of the economy to a shock of interest. The definition of "state-dependent" concerns the regime classification at the time at which the shock occurs. That is, the response of the economy to a shock occurring at time t is defined to be in the first state if the state variable, $y_{n,t-d}$, at time t, is below the threshold y_n^*, that is, $y_{n,t-d} \leq y_n^*$. Vice versa, the response of the economy to an identical shock occurring at time t' is defined to be in the second state if the state variable, $y_{n,t'-d}$, at time t', is above the threshold y_n^*, that is, $y_{n,t'-d} > y_n^*$. Importantly, after the shock hits, the state variable may respond endogenously and cross the threshold y_n^*, causing the economy to transition to the other regime. This does not alter the classification of the response: what defines the state is solely the position of $y_{n,t-d}$ relative to y_n^* at the moment the shock occurs, and what happens afterward constitutes a consequence of shock propagation and does not affect the original classification.

To estimate these objects, we use the Generalized Impulse Response Functions (GIRFs; Koop et al. 1996). This tool, as argued by Francis et al. (2023) and Gonçalves et al. (2024), accounts for endogenous transitions across states induced by the shock, thereby avoiding the potential bias of local projections, which instead capture average effects across regimes.

History-specific GIRFs are defined as the difference in expected dynamics between a treated scenario (shock $\mu = \delta$ at t) and an untreated scenario (shock $\mu = 0$ at t) conditional on a specific starting point $\mathbf{y}_{t-1} \equiv (Y_{t-p}, \ldots, Y_{t-1})$:

$$\text{GIRF}_t^h(\boldsymbol{\Psi}, \delta) \;\equiv\; \mathbb{E}[Y_{t+h} \mid \boldsymbol{\Psi},\, \mathbf{y}_{t-1},\, \mu = \delta] - \mathbb{E}[Y_{t+h} \mid \boldsymbol{\Psi},\, \mathbf{y}_{t-1},\, \mu = 0]\,, \tag{B.13}$$

where $\mathbb{E}[\cdot]$ denotes expectations over future innovations, $\boldsymbol{\Psi} \equiv (\mathbf{B}_1, \mathbf{B}_2, A_1, A_2, y_n^*, d)$ collects all model parameters; and A_i is the $n \times n$ structural impact matrix for regime i, satisfying $A_i A_i' = \Sigma_i$, with first column proportional to the regime-specific impact vector $\hat{\gamma}_i$ estimated by the first-stage regression (3), as defined in the main text. Each history-specific $\text{GIRF}_t^h(\boldsymbol{\Psi}, \delta)$, associated with

history $\mathbf{y}_{t-1}$, is then assigned to the first-regime group if $y_{n,t-d} \leq y_n^*$ and to the second-regime group if $y_{n,t-d} > y_n^*$. The state-i GIRF at horizon h is obtained by averaging all history-specific GIRFs in group i:

$$\mathrm{GIRF}_i^h(\boldsymbol{\Psi}, \delta) \;=\; \frac{1}{T_i} \sum_{t \in \mathcal{T}_i} \mathrm{GIRF}_t^h(\boldsymbol{\Psi}, \delta). \tag{B.14}$$

Simulation algorithm. For each history $\mathbf{y}_{t-1}$ taken from the observed sample, the history-specific $\mathrm{GIRF}_t^h(\boldsymbol{\Psi}, \delta)$ in (B.13) is approximated by Monte Carlo over R independent repetitions. In each repetition r, two parallel paths are simulated: a treated path $\{\hat{Y}_{t+h}^{T,r}\}_{h=0}^H$ and an untreated path $\{\hat{Y}_{t+h}^{U,r}\}_{h=0}^H$, both initialized at $\mathbf{y}_{t-1}$.

At $h = 0$ (impact), the treated path receives a structural shock of interest of size δ in the equation for the variable of interest, while the untreated path receives no shock. The regime is determined by $y_{n,t-d} \leq y_n^*$, evaluated on the actual data. For all $h \geq 1$, both paths draw independent standard normal innovations and propagate forward according to the threshold VAR:

$$\hat{Y}_{t+h}^{T,r} = S_{t+h}^T \left(\hat{X}_{t+h}^T \, \mathbf{B}_1 + \varepsilon_{t+h}^{T,r} \, A_1 \right) + \left(1 - S_{t+h}^T\right) \left(\hat{X}_{t+h}^T \, \mathbf{B}_2 + \varepsilon_{t+h}^{T,r} \, A_2 \right), \tag{B.15}$$

$$\hat{Y}_{t+h}^{U,r} = S_{t+h}^U \left(\hat{X}_{t+h}^U \, \mathbf{B}_1 + \varepsilon_{t+h}^{U,r} \, A_1 \right) + \left(1 - S_{t+h}^U\right) \left(\hat{X}_{t+h}^U \, \mathbf{B}_2 + \varepsilon_{t+h}^{U,r} \, A_2 \right), \tag{B.16}$$

where $S_{t+h}^T = 1$ if $\hat{y}_{n,t+h-d}^T \leq y_n^*$ and 0 otherwise, and $S_{t+h}^U = 1$ if $\hat{y}_{n,t+h-d}^U \leq y_n^*$ and 0 otherwise, with both regime indicators determined endogenously by the simulated path of the threshold variable in each respective path. The history-specific GIRF is then approximated as:

$$\mathrm{GIRF}_t^h(\boldsymbol{\Psi}, \delta) \;\approx\; \frac{1}{R} \sum_{r=1}^{R} \left(\hat{Y}_{t+h}^{T,r} - \hat{Y}_{t+h}^{U,r} \right), \tag{B.17}$$

and $\mathrm{GIRF}_i^h(\boldsymbol{\Psi}, \delta)$ is obtained by averaging over the relevant histories as in (B.14).

Posterior inference and normalization. The procedure above is repeated for each of the $(M_{\text{draws}} - M_{\text{burn}})/k$ retained posterior draws $\boldsymbol{\Psi}^{(m)}$, $m = 1, \ldots, M_{\text{store}}$: for each draw, the R-repetition Monte Carlo average is computed for every history in the observed sample, histories are assigned to their regime group as in (B.14), and averaging over $\mathcal{T}_1$ and $\mathcal{T}_2$ yields $\mathrm{GIRF}_1^h(\boldsymbol{\Psi}^{(m)}, \delta)$ and $\mathrm{GIRF}_2^h(\boldsymbol{\Psi}^{(m)}, \delta)$. The collection $\{\mathrm{GIRF}_i^h(\boldsymbol{\Psi}^{(m)}, \delta)\}_{m=1}^{M_{\text{store}}}$ constitutes the posterior distribution

of the state-i GIRF at horizon h. Posterior medians and 68% and 90% credible bands are computed as quantiles of this distribution.

C State-dependent local projections

As a further robustness check on the aggregate results of Section 2, we estimate the state-dependent effects of a monetary policy shock using a reduced-form local projection, following Jordà (2005) and, in its Bayesian formulation, Miranda-Agrippino and Ricco (2021).[14] We use this "reduced-form" local-projection approach rather than a "conventional" specification, as the latter would restrict the entire estimation to the shorter period over which the Bauer and Swanson (2023) proxy is available. Separating reduced-form dynamics from structural identification allows us to estimate the former on the full sample and use the shorter overlapping sample only to recover the state-specific structural impact vectors.

State indicator. Recall $Y_t = [y_{1,t}, y_{2,t}, \dots, y_{n,t}]'$, the same $n \times 1$ vector of endogenous variables used in the threshold VAR of Appendix B.1, and let S_t denote the same regime indicator,

$$S_t = 1 \iff y_{n,t-d} \leq 0, \tag{C.1}$$

where $y_{n,t}$ is again the threshold variable, the last element of Y_t. Two details distinguish (C.1) from its threshold-VAR counterpart in Appendix B.1. First, the zero threshold follows naturally from the fact that $y_{n,t}$ is a detrended (cyclical) series, with $y_{n,t-d} \leq 0$ identifying periods below trend and $y_{n,t-d} > 0$ identifying periods above trend. Thus, unlike in the threshold VAR of Appendix B.1, the threshold is imposed deterministically rather than jointly estimated with the rest of the system. Second, the delay $d \in \{1, 2, 3, \dots\}$ is fixed at a chosen value rather than estimated jointly with the rest of the model.

Reduced-form coefficients. For each horizon $h = 0, 1, \dots, H$, we estimate the direct, multi-step analogue of the threshold VAR of Appendix B.1, in the same form but with every term indexed

[14] See Appendix B of the online appendix to Miranda-Agrippino and Ricco (2021) for a detailed treatment of the reduced-form local projection, in both its frequentist and Bayesian forms, for the linear case. Our extension follows the same reduced-form logic, generalized to allow for two regimes, but is estimated with the frequentist estimator throughout, without the Bayesian shrinkage of Miranda-Agrippino and Ricco (2021).

by h:

$$Y_{t+h} = \left[c_{1,h} + \sum_{j=1}^{p} B_{1,j}^{(h)} Y_{t-j} + u_{1,t+h}^{(h)} \right] S_t + \left[c_{2,h} + \sum_{j=1}^{p} B_{2,j}^{(h)} Y_{t-j} + u_{2,t+h}^{(h)} \right] (1 - S_t), \tag{C.2}$$

where $c_{i,h}$ is the horizon-h, regime-i intercept, $B_{i,j}^{(h)} \in \mathbb{R}^{n \times n}$ is the horizon-h, regime-i coefficient on lag Y_{t-j}, and $u_{i,t+h}^{(h)}$ is the associated projection residual. At $h = 0$, equation (C.2) coincides exactly with the threshold VAR itself, with $c_{i,0} \equiv c_i$, $B_{i,j}^{(0)} \equiv B_{i,j}$, and $u_{i,t}^{(0)} \equiv u_{i,t}$; for $h \geq 1$ it is its direct, h-step-ahead analogue. Equation (C.2) is estimated by ordinary least squares, equation by equation, separately for each $h = 0, 1, \ldots, H$.

Structural identification. We identify the state-specific structural impact vector using the residuals of the $h = 0$ regression, $u_{i,t}^{(0)} = u_{i,t}$, exactly as in equation (3) of the main text.

Impulse responses. The impact response is the structural vector itself, $\mathrm{IRF}_i^{LP}(0) = \gamma_i$. For $h = 1, \ldots, H$, the response h periods after impact is obtained by propagating γ_i forward through the coefficient that the $(h-1)$-th direct-horizon regression attaches to Y_{t-1}, its most recent regressor:

$$\mathrm{IRF}_i^{LP}(h) = \gamma_i \, B_{i,1}^{(h-1)}, \qquad h = 1, \ldots, H. \tag{C.3}$$

Inference. We conduct inference following the frequentist implementation of the reduced-form local projections in Miranda-Agrippino and Ricco (2021). In particular, we treat the first-stage structural impact vector γ_i as given, so that all sampling uncertainty arises from the estimation of $B_{i,1}^{(h)}$.[15] Because $B_{i,1}^{(h)}$ is estimated by ordinary least squares, we compute its covariance matrix using a Bartlett-kernel Newey–West heteroskedasticity- and autocorrelation-consistent estimator. Let $\Sigma_{B_{i,1},j}^{(h)}$ denote the resulting covariance matrix for the coefficients on Y_{t-1} that determine the response of variable j at horizon h. Since the impulse response $\mathrm{IRF}_{i,j}^{LP}(h) = \gamma_i B_{i,1}^{(h-1)}$ is a linear combination of these coefficients, with weights given by γ_i, its variance is

$$\mathrm{Var}\left(\mathrm{IRF}_{i,j}^{LP}(h)\right) = \gamma_i, \Sigma_{B_{i,1},j}^{(h-1)}, \gamma_i', \qquad h = 1, \ldots, H. \tag{C.4}$$

[15] At $h = 0$, $\mathrm{IRF}_i^{LP}(0) = \gamma_i$ implying no sample uncertainty on impact. This is consistent with Miranda-Agrippino and Ricco (2021)'s own frequentist local projection, which likewise reports no standard errors on the impact responses of the endogenous variables.

Difference in impulse responses across states. We additionally report the difference between the two regimes' impulse responses,

$$\Delta\text{IRF}_j^{LP}(h) \equiv \text{IRF}_{1,j}^{LP}(h) - \text{IRF}_{2,j}^{LP}(h), \qquad h = 0, 1, \ldots, H, \tag{C.5}$$

where regime 1 ($S_t = 1$) is the low-inventory state and regime 2 is the high-inventory state, consistent with the sign convention used throughout the paper. Because the regime-interacted regressors $S_t Y_{t-j}$ and $(1 - S_t)Y_{t-j}$ have non-overlapping support for every j, the two regimes' coefficients are estimated from disjoint observations and their sampling covariance is exactly zero, $\text{Cov}\left(B_{1,j}^{(h)}, B_{2,j}^{(h)}\right) = 0$ for every j and h. Consequently, the variance of the difference in (C.5) is simply the sum of the two regimes' own variances,

$$\text{Var}\big(\Delta\text{IRF}_j^{LP}(h)\big) = \text{Var}\big(\text{IRF}_{1,j}^{LP}(h)\big) + \text{Var}\big(\text{IRF}_{2,j}^{LP}(h)\big), \tag{C.6}$$

with no covariance correction required.[16]

[16] Equation (C.6) coincides with the variance that would be obtained from an algebraically equivalent reparametrization of (C.2), in which one regime's coefficients are replaced by a common baseline level and the other regime is captured by an interaction with a dummy variable: because the two parametrizations span the same column space, they deliver identical fitted values, residuals, and coefficient covariances, and (C.6) is simply the resulting variance of that dummy-interacted coefficient, expressed here in terms of the two regimes' separately estimated variances.

D Additional results

D.1 Aggregate evidence

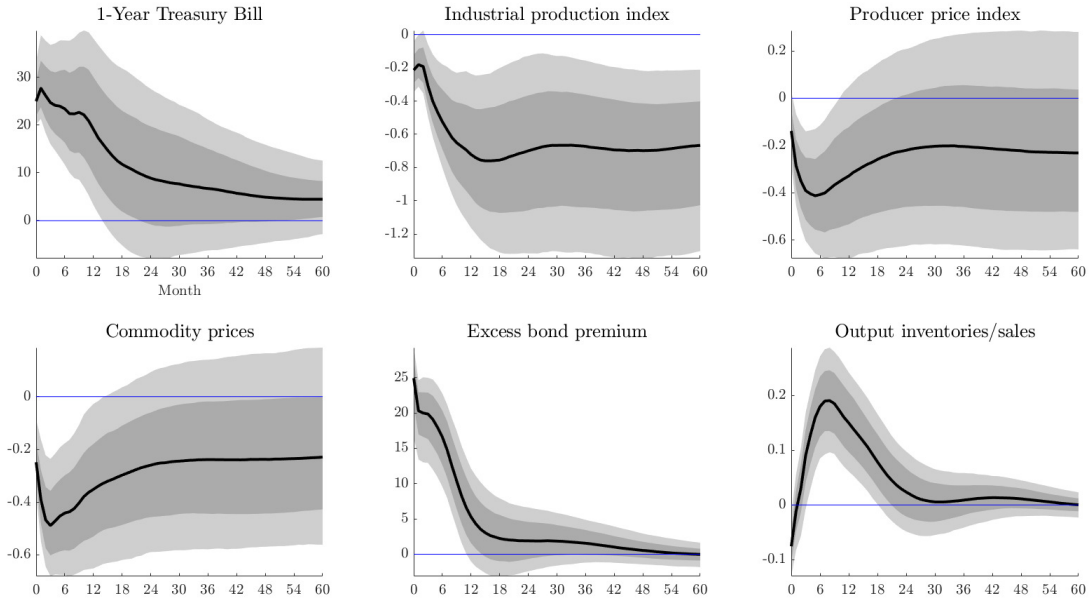


Figure D1: Linear Bayesian VAR

Note: Each panel shows the impulse response to a contractionary monetary policy shock, for the one-year Treasury bill rate (in basis points), the log of industrial production, the log of the PPI, the log of commodity prices, the excess bond premium (in basis points), and the filtered output inventory-to-sales ratio for the baseline specification. The solid line is the posterior median of the difference; dark and light shaded areas correspond to 68% and 90% posterior credible bands, respectively. The responses are normalized so that the impact effect of the shock on the one-year Treasury bill equals 25 basis points. The impact effect γ of a monetary policy shock is identified using the proxy of Bauer and Swanson (2023).

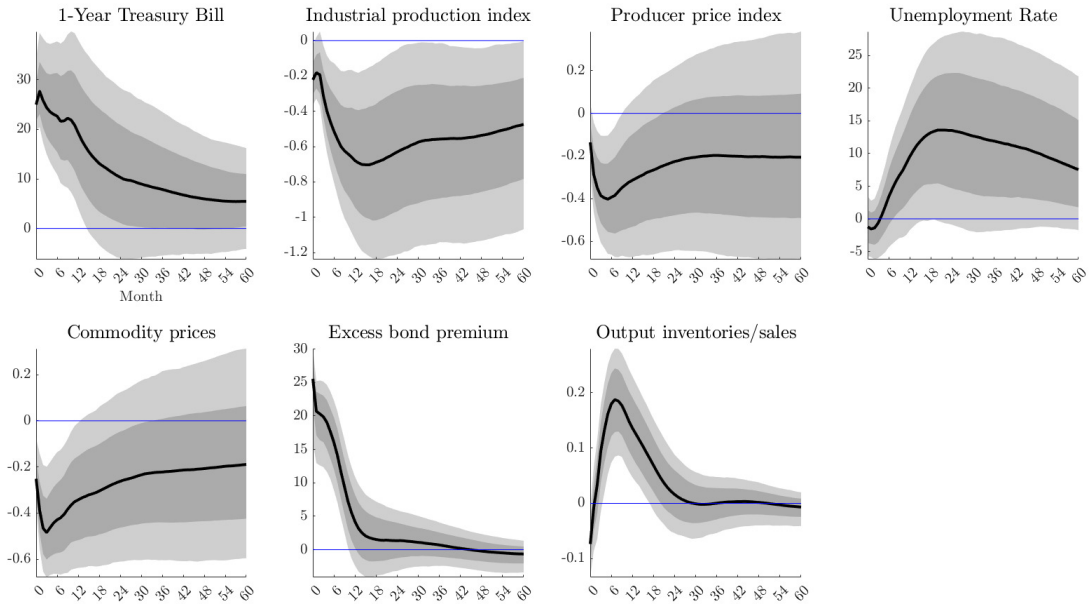


Figure D2: Linear Bayesian VAR with unemployment rate

Note: Each panel shows the impulse response to a contractionary monetary policy shock, for the one-year Treasury bill rate (in basis points), the log of industrial production, the log of the PPI, the unemployment rate (in basis points), the log of commodity prices, the excess bond premium (in basis points), and the filtered output inventory-to-sales ratio for the baseline specification. The solid line is the posterior median of the difference; dark and light shaded areas correspond to 68% and 90% posterior credible bands, respectively. The responses are normalized so that the impact effect of the shock on the one-year Treasury bill equals 25 basis points. The impact effect γ of a monetary policy shock is identified using the proxy of Bauer and Swanson (2023).

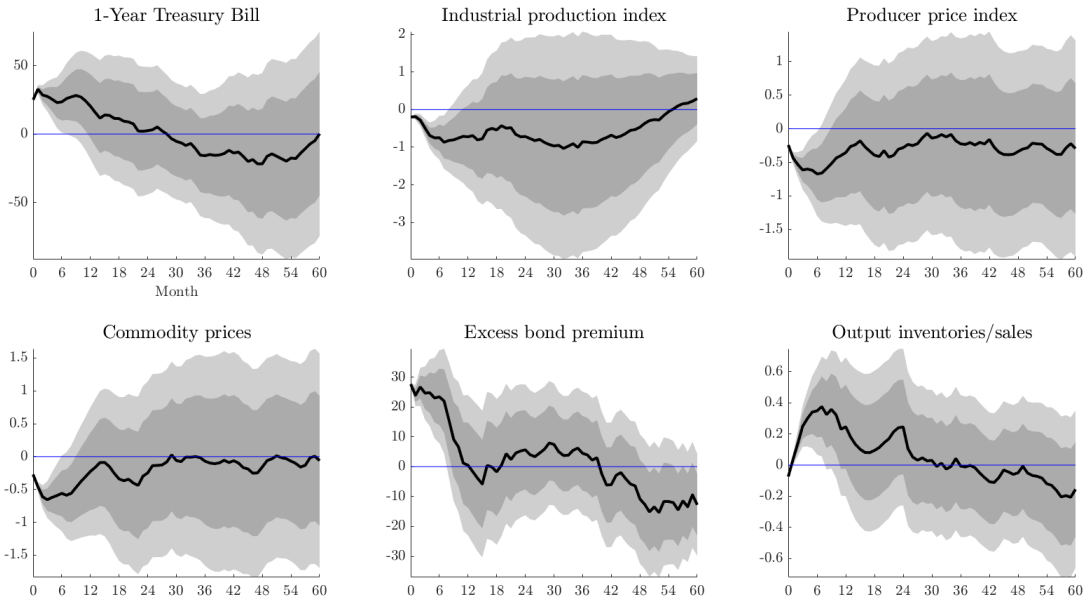
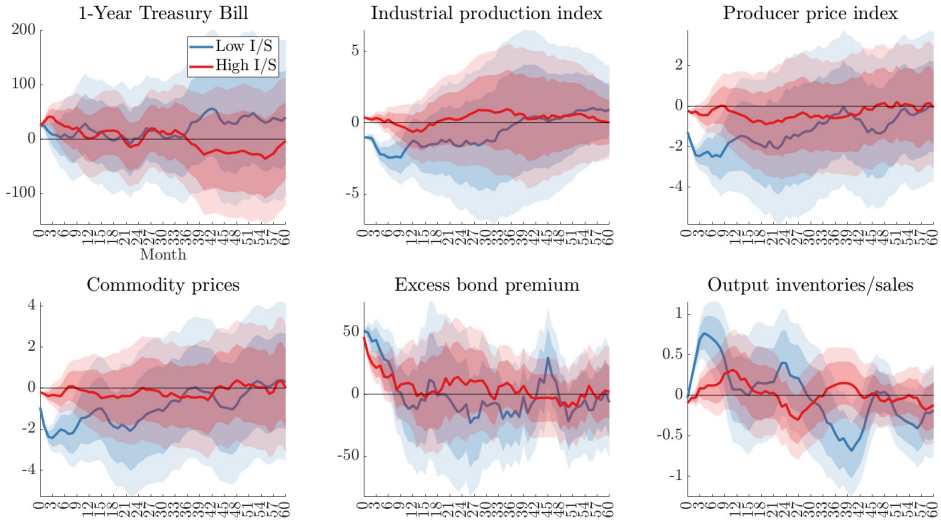
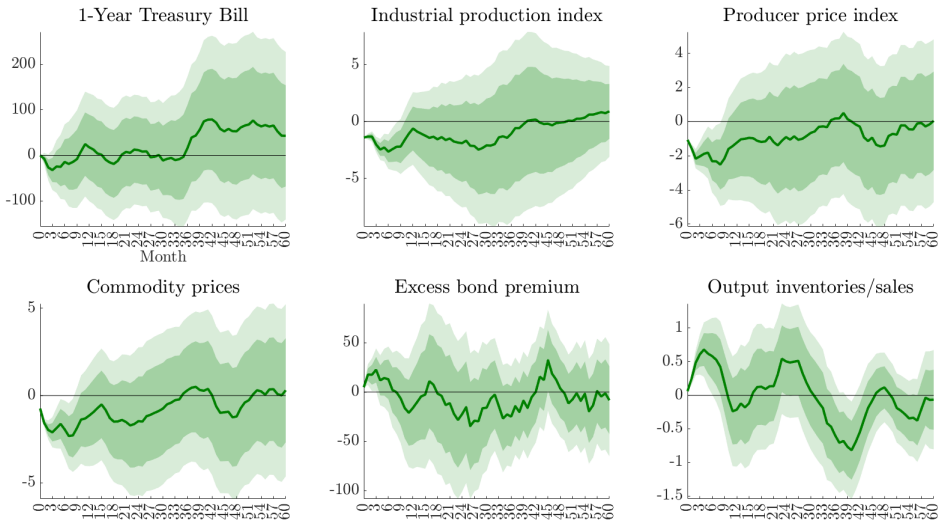


Figure D3: Linear local projections

Note: Each panel shows the impulse response to a contractionary monetary policy shock, for the one-year Treasury bill rate (in basis points), the log of industrial production, the log of the PPI, the log of commodity prices, the excess bond premium (in basis points), and the filtered output inventory-to-sales ratio for the baseline specification. The solid line is the point estimate; dark and light shaded areas correspond to 68% and 90% confidence bands, respectively. The responses are normalized so that the impact effect of the shock on the one-year Treasury bill equals 25 basis points. The impact effect γ of a monetary policy shock is identified using the proxy of Bauer and Swanson (2023).



(a) State-dependent IRFs to a monetary policy shock



(b) Difference in state-dependent IRFs

Figure D4: Local projections – Longer horizon

Note: Panel (a) shows the state-dependent IRFs; Panel (b) shows the difference between the low- and the high-inventory regime responses. Each panel reports the response of the endogenous variables to a monetary policy shock, normalized so that the impact effect on the Treasury bill equals 25bp. Response of the Treasury bill and EBP are in basis point, while the response of the remain variables are in percent. Solid lines are point estimates; dark and light shaded areas correspond to 68% and 90% confidence bands, respectively; blue (red) lines and bands correspond to the low- (high-) inventory regime. The effects of the shock are identified using the proxy of Bauer and Swanson (2023).

D.2 Sector-level evidence

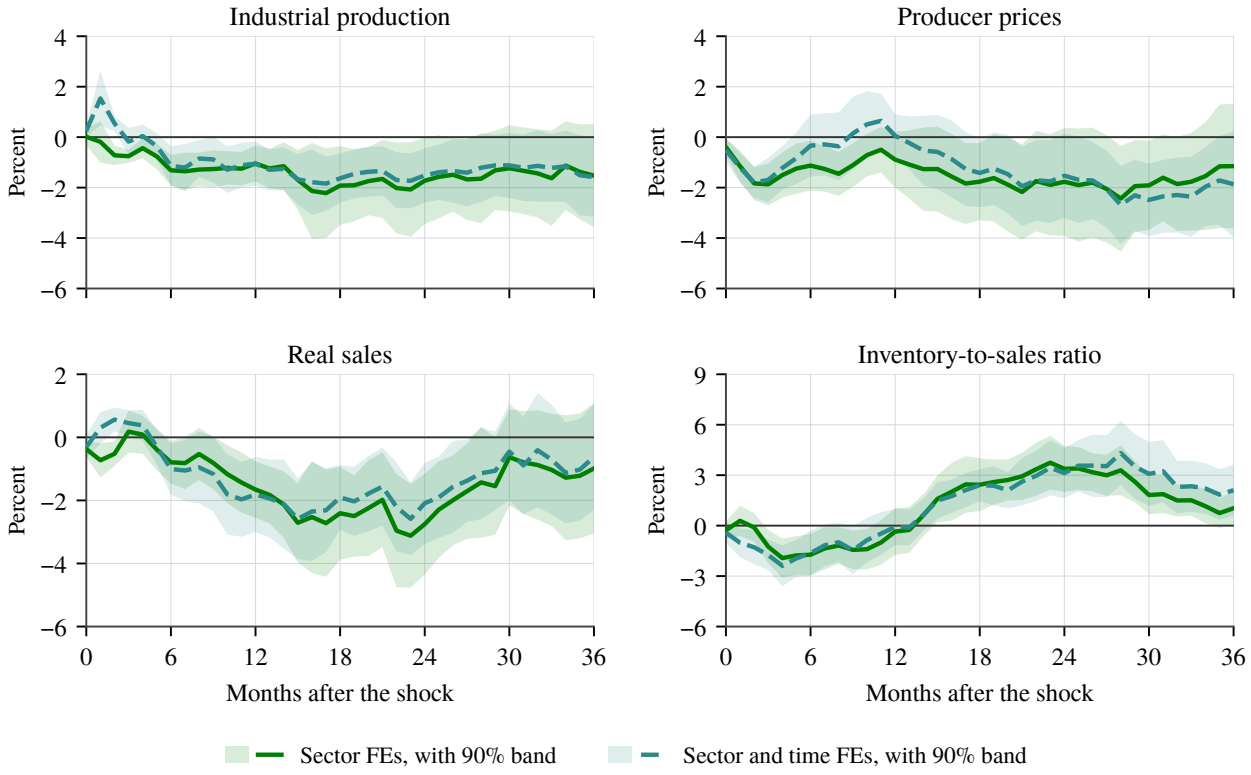


Figure D5: Effect of the Continuous Inventory State

Note: This figure presents the effect of the continuous inventory state based on the sector-level log IS ratio, detrended by the same high-pass filter as in the aggregate analysis to remove each sector's long-term trend beyond the periodicity of 10 years. We plot the impulse responses after replacing $S_{j,t-1}$ in Equation (4) with the negative of the continuous state to show the effect of lower inventories. Consistent with Figure 5, real sales are deflated by the sector's own producer price index. Shaded areas are 90% confidence intervals based on Driscoll–Kraay standard errors to account for correlations across sectors and time, where the maximum lag order is set to the maximum projection horizon. Bauer and Swanson monetary policy shocks are normalized to raise the one-year Treasury rate by 25 basis points on impact.

E Robustness

E.1 Filter

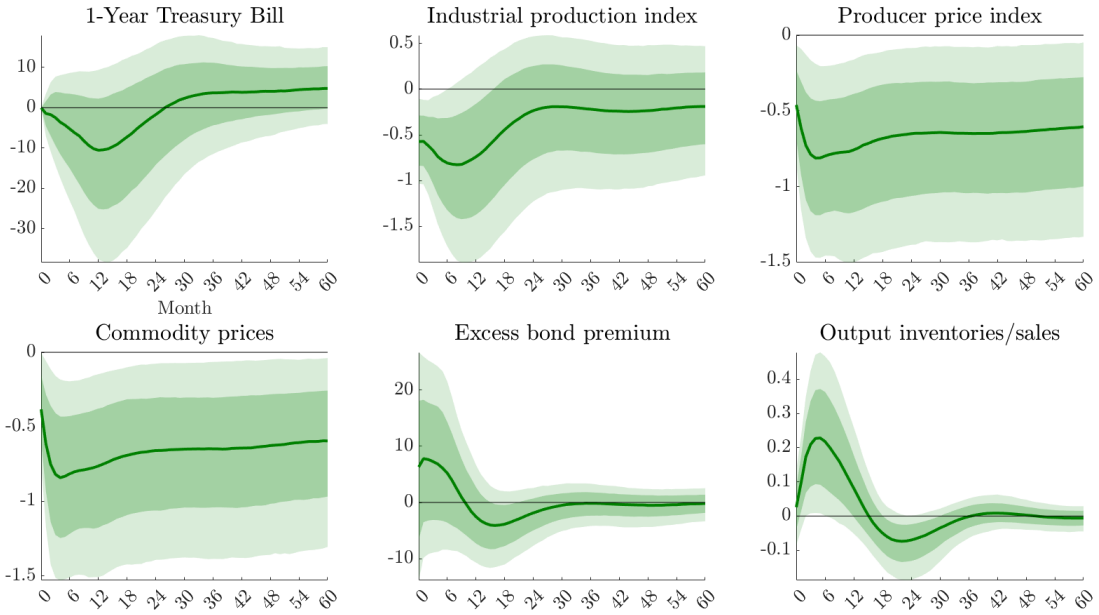


Figure E1: Hodrick and Prescott (1997) filter for Inventories/sales – Difference in GIRFs

Note: Each panel shows the difference between the low- and the high-inventory regime generalized impulse response to a contractionary monetary policy shock, for the one-year Treasury bill rate (in basis points), the log of industrial production, the log of the PPI, the log of commodity prices, the excess bond premium (in basis points), and the output inventory-to-sales ratio, detrended with the Hodrick and Prescott (1997) filter. A negative value indicates a more negative response in the low-inventory regime. The solid line is the posterior median of the difference; dark and light shaded areas correspond to 68% and 90% posterior credible bands, respectively. Responses are expressed in the same units as Figure 2a: the shock is normalized so that the on-impact effect on the one-year Treasury bill equals 25 basis points. The state-dependent impact effect γ_i of a monetary policy shock is identified using the proxy of Bauer and Swanson (2023). See Appendix B.1 for the structural identification and Appendix B.2 for the construction of the generalized impulse response functions.

E.2 Monetary policy shock series

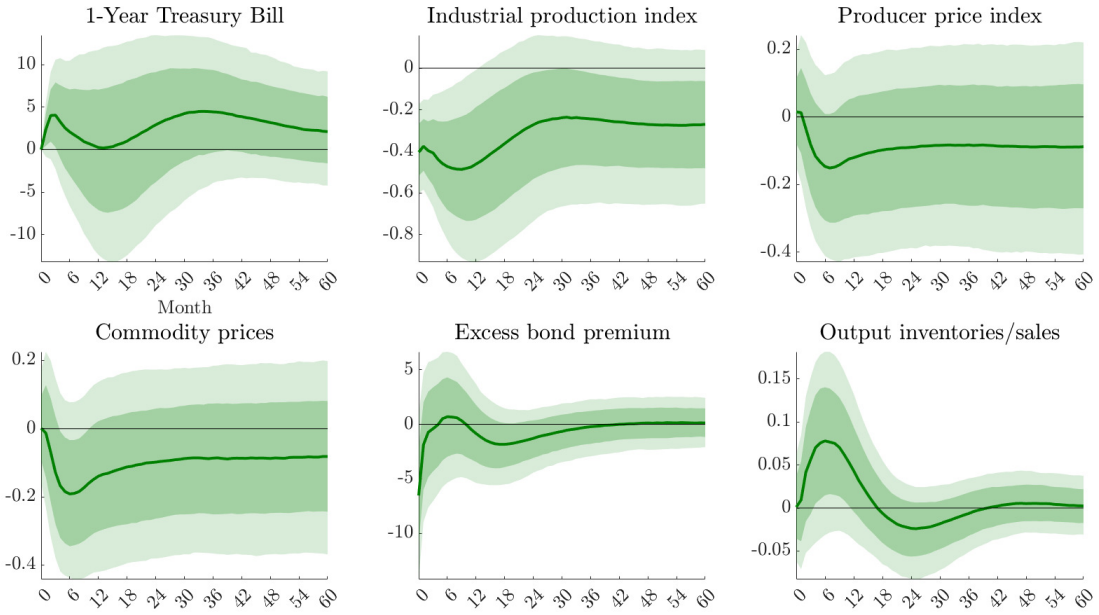


Figure E2: MP shocks by Miranda-Agrippino and Ricco (2021) – Difference in GIRFs

Note: Each panel shows the difference between the low- and the high-inventory regime generalized impulse response to a contractionary monetary policy shock, for the one-year Treasury bill rate (in basis points), the log of industrial production, the log of the PPI, the log of commodity prices, the excess bond premium (in basis points), and the filtered output inventory-to-sales ratio. A negative value indicates a more negative response in the low-inventory regime. The solid line is the posterior median of the difference; dark and light shaded areas correspond to 68% and 90% posterior credible bands, respectively. Responses are expressed in the same units as Figure 2a: the shock is normalized so that the on-impact effect on the one-year Treasury bill equals 25 basis points. The state-dependent impact effect γ_i of a monetary policy shock is identified using the proxy of Miranda-Agrippino and Ricco (2021) extended by Degasperi and Ricco (2024). See Appendix B.1 for the structural identification and Appendix B.2 for the construction of the generalized impulse response functions.

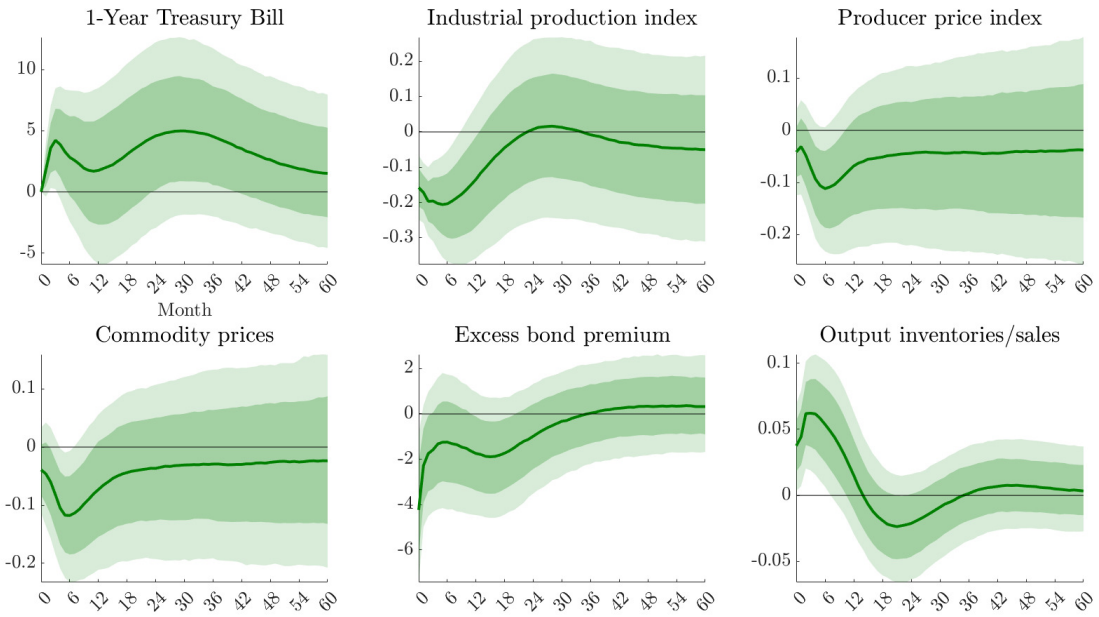


Figure E3: MP shocks by Aruoba and Drechsel (2024) – Difference in GIRFs

Note: Each panel shows the difference between the low- and the high-inventory regime generalized impulse response to a contractionary monetary policy shock, for the one-year Treasury bill rate (in basis points), the log of industrial production, the log of the PPI, the log of commodity prices, the excess bond premium (in basis points), and the filtered output inventory-to-sales ratio. A negative value indicates a more negative response in the low-inventory regime. The solid line is the posterior median of the difference; dark and light shaded areas correspond to 68% and 90% posterior credible bands, respectively. Responses are expressed in the same units as Figure 2a: the shock is normalized so that the on-impact effect on the one-year Treasury bill equals 25 basis points. The state-dependent impact effect γ_i of a monetary policy shock is identified using the proxy of Aruoba and Drechsel (2024). See Appendix B.1 for the structural identification and Appendix B.2 for the construction of the generalized impulse response functions.

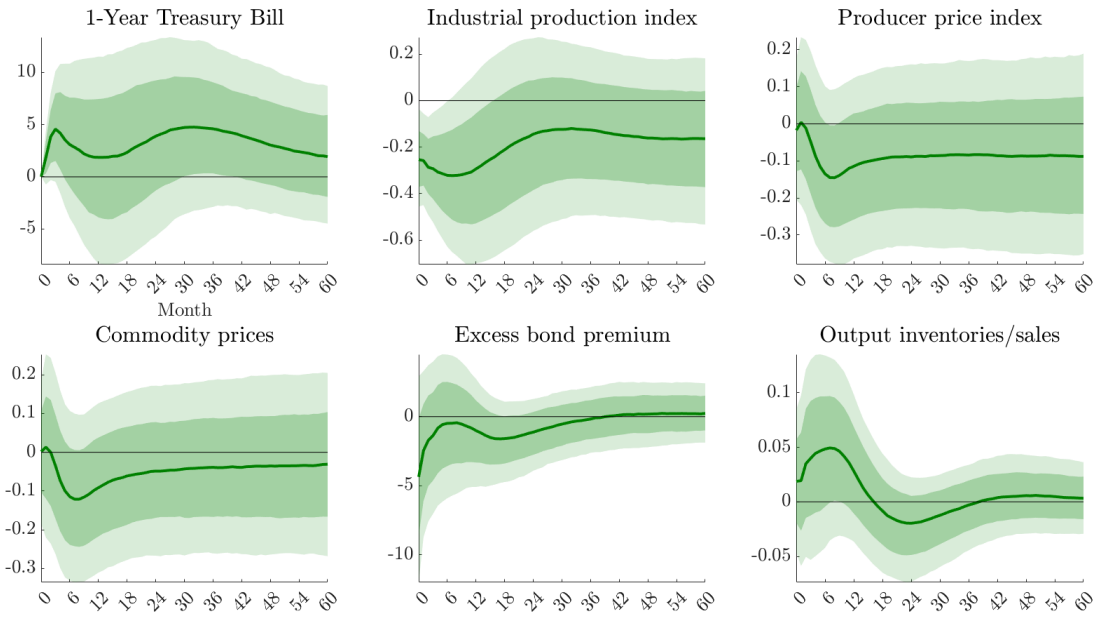


Figure E4: MP shocks by Romer and Romer (2004) – Difference in GIRFs

Note: Each panel shows the difference between the low- and the high-inventory regime generalized impulse response to a contractionary monetary policy shock, for the one-year Treasury bill rate (in basis points), the log of industrial production, the log of the PPI, the log of commodity prices, the excess bond premium (in basis points), and the filtered output inventory-to-sales ratio. A negative value indicates a more negative response in the low-inventory regime. The solid line is the posterior median of the difference; dark and light shaded areas correspond to 68% and 90% posterior credible bands, respectively. Responses are expressed in the same units as Figure 2a: the shock is normalized so that the on-impact effect on the one-year Treasury bill equals 25 basis points. The state-dependent impact effect γ_i of a monetary policy shock is identified using the proxy of Romer and Romer (2004) extended by Wieland (2021). See Appendix B.1 for the structural identification and Appendix B.2 for the construction of the generalized impulse response functions.

E.3 Inventory measure

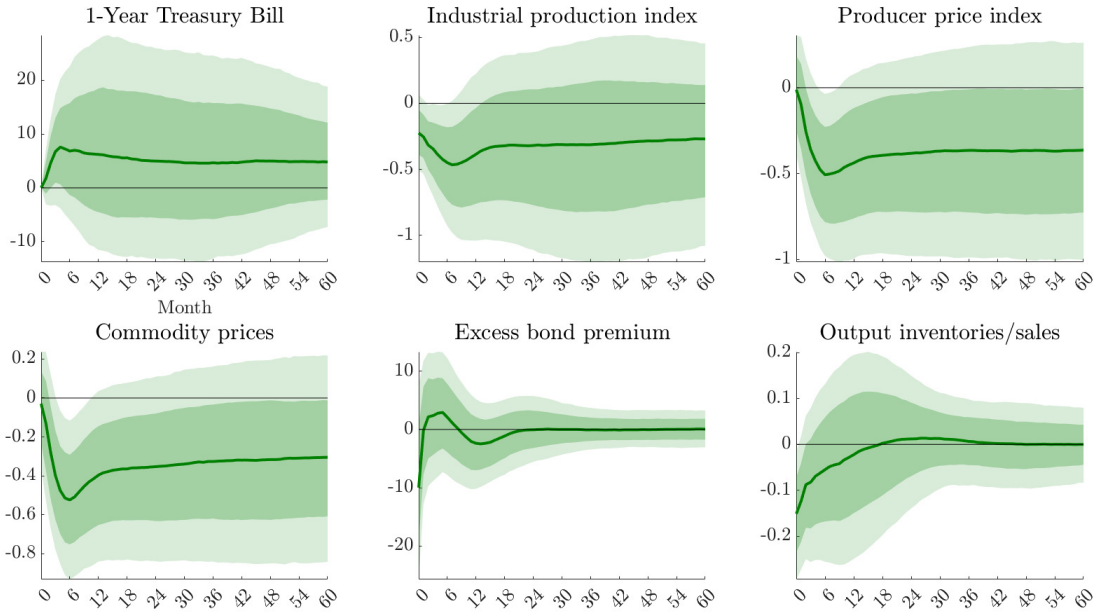


Figure E5: Removing backlogs of orders – Difference in GIRFs

Note: Each panel shows the difference between the low- and the high-inventory regime generalized impulse response to a contractionary monetary policy shock, for the one-year Treasury bill rate (in basis points), the log of industrial production, the log of the PPI, the log of commodity prices, the excess bond premium (in basis points), and the filtered output inventory-to-sales ratio, with inventories measured net of backlog orders, for the baseline specification. A negative value indicates a more negative response in the low-inventory regime. The solid line is the posterior median of the difference; dark and light shaded areas correspond to 68% and 90% posterior credible bands, respectively. Responses are expressed in the same units as Figure 2a: the shock is normalized so that the on-impact effect on the one-year Treasury bill equals 25 basis points. The state-dependent impact effect γ_i of a monetary policy shock is identified using the proxy of Bauer and Swanson (2023). See Appendix B.1 for the structural identification and Appendix B.2 for the construction of the generalized impulse response functions.

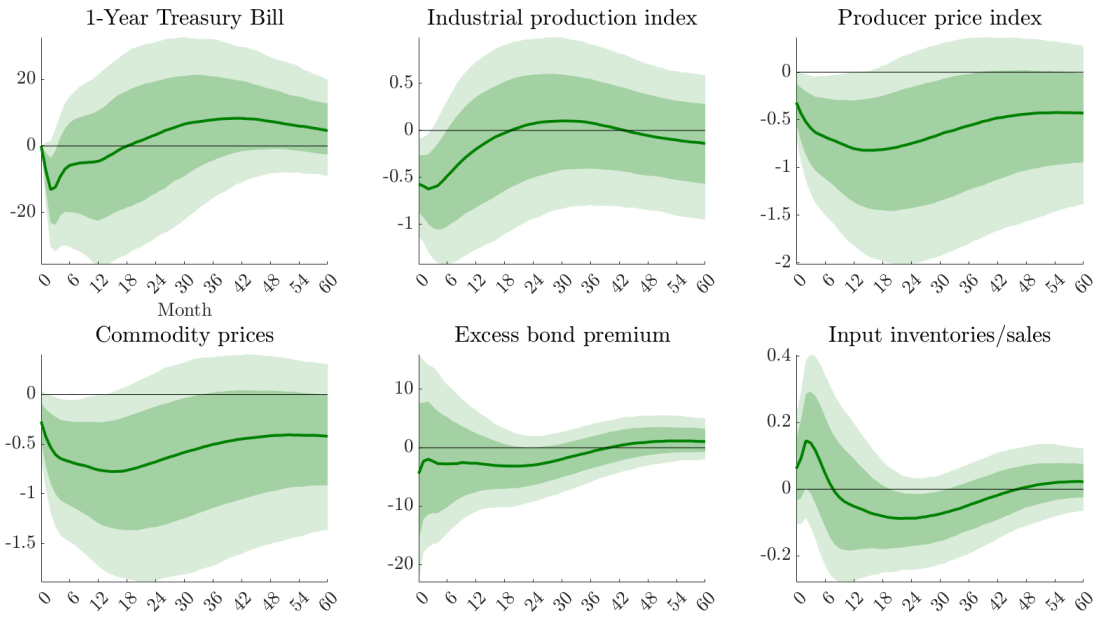


Figure E6: Input inventories – Difference in GIRFs

Note: Each panel shows the difference between the low- and the high-inventory regime generalized impulse response to a contractionary monetary policy shock, for the one-year Treasury bill rate (in basis points), the log of industrial production, the log of the PPI, the log of commodity prices, the excess bond premium (in basis points), and the filtered input inventory-to-sales ratio for the baseline specification. A negative value indicates a more negative response in the low-inventory regime. The solid line is the posterior median of the difference; dark and light shaded areas correspond to 68% and 90% posterior credible bands, respectively. Responses are expressed in the same units as Figure 2a: the shock is normalized so that the on-impact effect on the one-year Treasury bill equals 25 basis points. The state-dependent impact effect γ_i of a monetary policy shock is identified using the proxy of Bauer and Swanson (2023). See Appendix B.1 for the structural identification and Appendix B.2 for the construction of the generalized impulse response functions.

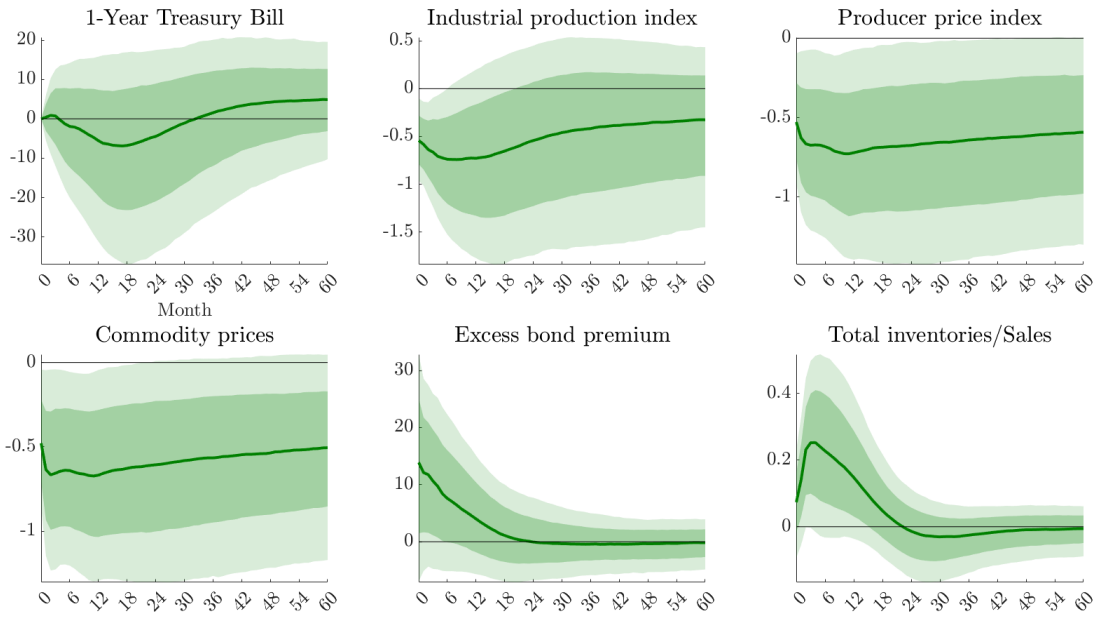


Figure E7: Total inventories – Difference in GIRFs

Note: Each panel shows the difference between the low- and the high-inventory regime generalized impulse response to a contractionary monetary policy shock, for the one-year Treasury bill rate (in basis points), the log of industrial production, the log of the PPI, the log of commodity prices, the excess bond premium (in basis points), and the filtered total inventory-to-sales ratio. A negative value indicates a more negative response in the low-inventory regime. The solid line is the posterior median of the difference; dark and light shaded areas correspond to 68% and 90% posterior credible bands, respectively. Responses are expressed in the same units as Figure 2a: the shock is normalized so that the on-impact effect on the one-year Treasury bill equals 25 basis points. The state-dependent impact effect γ_i of a monetary policy shock is identified using the proxy of Bauer and Swanson (2023). See Appendix B.1 for the structural identification and Appendix B.2 for the construction of the generalized impulse response functions.

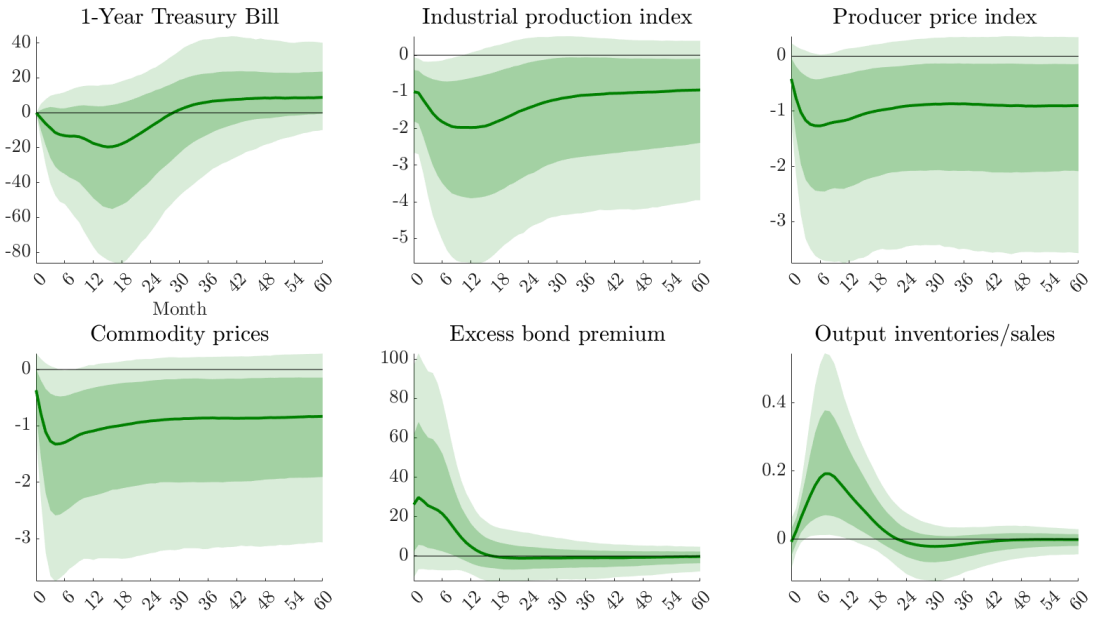


Figure E8: Sales over the last two quarters as denominator – Difference in GIRFs

Note: Each panel shows the difference between the low- and the high-inventory regime generalized impulse response to a contractionary monetary policy shock, for the one-year Treasury bill rate (in basis points), the log of industrial production, the log of the PPI, the log of commodity prices, the excess bond premium (in basis points), and the filtered output inventory-to-sales ratio, where sales are measured as cumulative sales over the previous two quarters. A negative value indicates a more negative response in the low-inventory regime. The solid line is the posterior median of the difference; dark and light shaded areas correspond to 68% and 90% posterior credible bands, respectively. Responses are expressed in the same units as Figure 2a: the shock is normalized so that the on-impact effect on the one-year Treasury bill equals 25 basis points. The state-dependent impact effect γ_i of a monetary policy shock is identified using the proxy of Bauer and Swanson (2023). See Appendix B.1 for the structural identification and Appendix B.2 for the construction of the generalized impulse response functions.

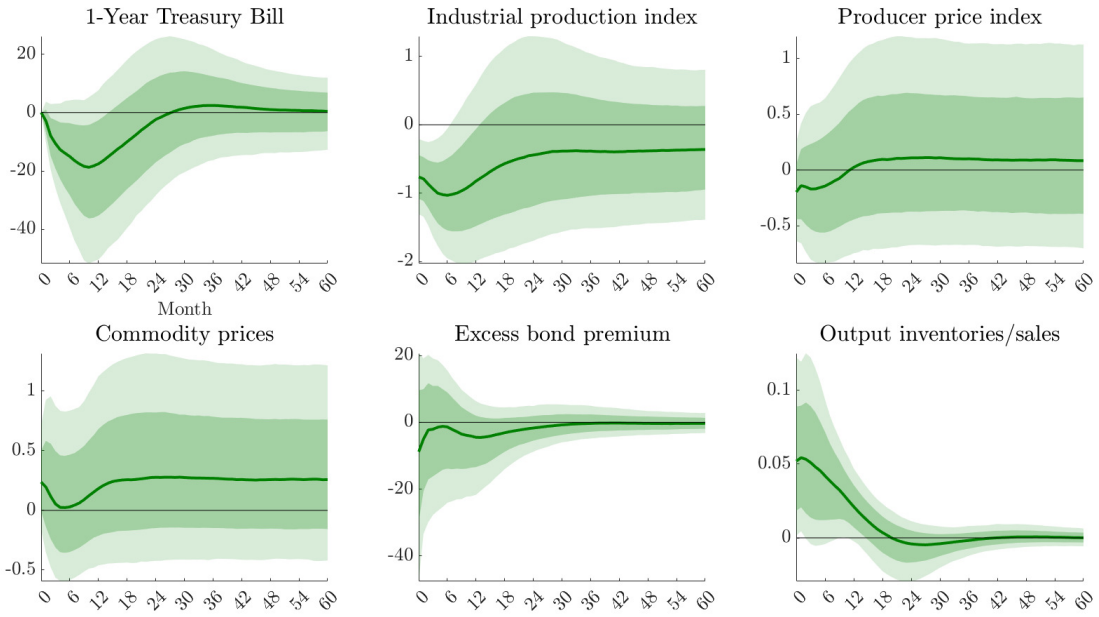


Figure E9: Sales over the last year as denominator – Difference in GIRFs

Note: Each panel shows the difference between the low- and the high-inventory regime generalized impulse response to a contractionary monetary policy shock, for the one-year Treasury bill rate (in basis points), the log of industrial production, the log of the PPI, the log of commodity prices, the excess bond premium (in basis points), and the filtered output inventory-to-sales ratio, where sales are measured as cumulative sales over the previous year. A negative value indicates a more negative response in the low-inventory regime. The solid line is the posterior median of the difference; dark and light shaded areas correspond to 68% and 90% posterior credible bands, respectively. Responses are expressed in the same units as Figure 2a: the shock is normalized so that the on-impact effect on the one-year Treasury bill equals 25 basis points. The state-dependent impact effect γ_i of a monetary policy shock is identified using the proxy of Bauer and Swanson (2023). See Appendix B.1 for the structural identification and Appendix B.2 for the construction of the generalized impulse response functions.

E.4 Additional variables

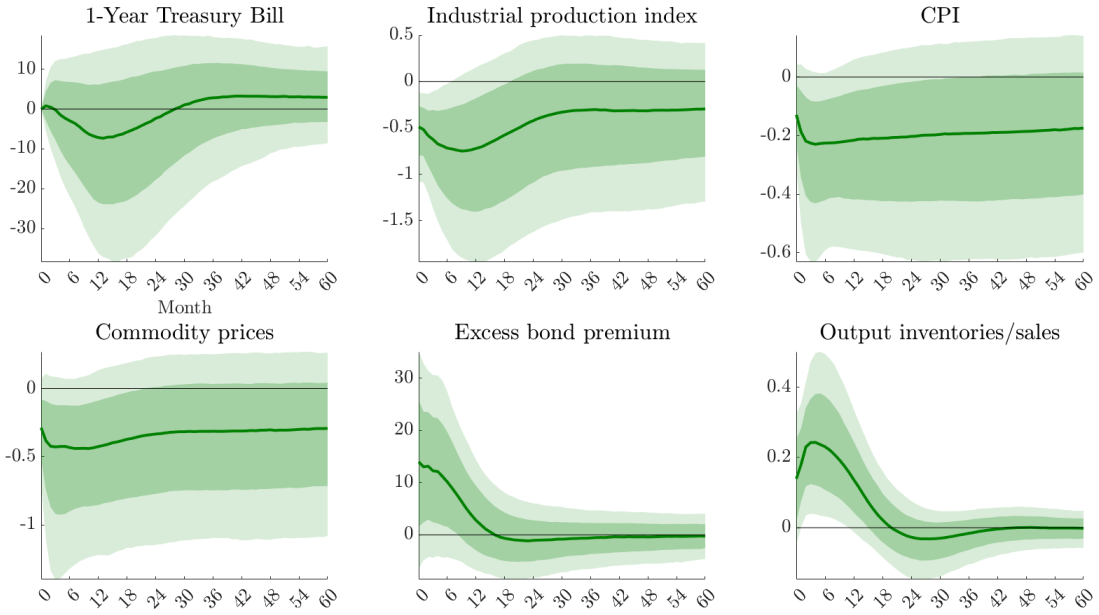


Figure E10: Adding consumer price index (CPI) – Difference in GIRFs

Note: Each panel shows the difference between the low- and the high-inventory regime generalized impulse response to a contractionary monetary policy shock, for the one-year Treasury bill rate (in basis points), the log of industrial production, the log of the CPI, the log of commodity prices, the excess bond premium (in basis points), and the filtered input inventory-to-sales ratio for the baseline specification. A negative value indicates a more negative response in the low-inventory regime. The solid line is the posterior median of the difference; dark and light shaded areas correspond to 68% and 90% posterior credible bands, respectively. Responses are expressed in the same units as Figure 2a: the shock is normalized so that the on-impact effect on the one-year Treasury bill equals 25 basis points. The state-dependent impact effect γ_i of a monetary policy shock is identified using the proxy of Bauer and Swanson (2023). See Appendix B.1 for the structural identification and Appendix B.2 for the construction of the generalized impulse response functions.

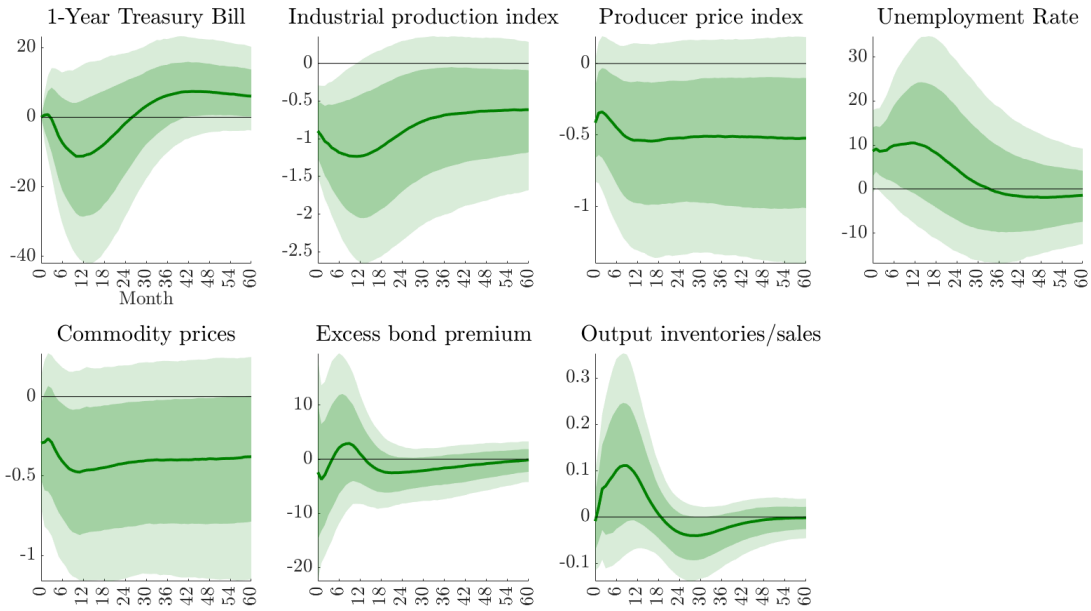


Figure E11: Adding unemployment rate – Difference in GIRFs

Note: Each panel shows the difference between the low- and the high-inventory regime generalized impulse response to a contractionary monetary policy shock, for the one-year Treasury bill rate (in basis points), the log of industrial production, the log of the PPI, the unemployment rate (in basis points), the log of commodity prices, the excess bond premium (in basis points), and the filtered input inventory-to-sales ratio for the baseline specification. A negative value indicates a more negative response in the low-inventory regime. The solid line is the posterior median of the difference; dark and light shaded areas correspond to 68% and 90% posterior credible bands, respectively. Responses are expressed in the same units as Figure 2a: the shock is normalized so that the on-impact effect on the one-year Treasury bill equals 25 basis points. The state-dependent impact effect γ_i of a monetary policy shock is identified using the proxy of Bauer and Swanson (2023). See Appendix B.1 for the structural identification and Appendix B.2 for the construction of the generalized impulse response functions.

F Model

F.1 Proofs

Proof of Proposition 1. The payoff $(1-\delta)^{\tau} M_{t,t+\tau}$ is strictly decreasing in τ under the stated condition. Raising a hazard rate shifts the distribution of τ down in the sense of first-order stochastic dominance, and the expectation of a strictly decreasing function of τ therefore rises. ■

Proof of Proposition 2. Divide (10) by x and use (13) and (12) to obtain the first equality in (35). For the second, the lognormal partial expectation with $\mathrm{E}\varepsilon = 1$ gives $\mathrm{E}[\varepsilon \mathbf{1}\{\varepsilon < k\}] = \mathcal{N}(d_1)$, so the numerator $xS_x = x\mathcal{N}(d_1)$ is expected sales at outlets whose demand falls short of the cap.

Since $p = 1 - \mathcal{N}(d_0)$ is a strictly decreasing bijection from $d_0 \in \mathbb{R}$ onto $(0,1)$, it suffices to show that ϵ^{eff} is strictly increasing in d_0. Write $\epsilon^{\text{eff}}/\theta = A/(A+C)$ with

$$A \equiv \mathcal{N}(d_1), \qquad C \equiv k\left[1 - \mathcal{N}(d_0)\right] = e^{\sigma_d d_0 - \sigma_d^2/2}\left[1 - \mathcal{N}(d_0)\right], \tag{F.1}$$

both understood as functions of d_0 alone, with $A + C = S/x$. Because $d_0 = d_1 + \sigma_d$ and $\sigma_d d_1 = \log k - \sigma_d^2/2$, the two densities satisfy

$$\phi_N(d_0) = \phi_N(d_1) e^{-\sigma_d d_1 - \sigma_d^2/2} = \frac{\phi_N(d_1)}{k}. \tag{F.2}$$

Differentiating and using (F.2),

$$A' = \phi_N(d_1), \qquad C' = \sigma_d C - k\phi_N(d_0) = \sigma_d C - \phi_N(d_1), \tag{F.3}$$

so that

$$\frac{d}{dd_0}\frac{A}{A+C} = \frac{A'C - AC'}{(A+C)^2} = \frac{\phi_N(d_1)(A+C) - \sigma_d AC}{(A+C)^2}. \tag{F.4}$$

Divide the numerator by $AC > 0$. Since $\phi_N(d_1)/A = \phi_N(d_1)/\mathcal{N}(d_1)$ and, by (F.2) once more, $\phi_N(d_1)/C = \phi_N(d_0)/[1 - \mathcal{N}(d_0)]$,

$$\operatorname{sign}\frac{d\epsilon^{\text{eff}}}{dd_0} = \operatorname{sign}\left[h(d_0) + h(-d_1) - \sigma_d\right], \qquad h(y) \equiv \frac{\phi_N(y)}{1 - \mathcal{N}(y)}. \tag{F.5}$$

The normal hazard satisfies $h(y) > y$ for all $y \in \mathbb{R}$: this is immediate for $y \leq 0$, and for $y > 0$ it follows from $\int_y^\infty \phi_N(u)du < \int_y^\infty (u/y)\phi_N(u)du = \phi_N(y)/y$. Applying the bound to each term,

$$h(d_0) + h(-d_1) > d_0 - d_1 = \sigma_d, \tag{F.6}$$

so the bracket in (F.5) is strictly positive for every d_0 and every $\sigma_d > 0$. For the limits, $p \to 0$ sends $d_0 \to \infty$, where $A \to 1$ and $C \to 0$ because $1 - \mathcal{N}(d_0) < \phi_N(d_0)/d_0$; and $p \to 1$ sends $d_0 \to -\infty$, where $A = \mathcal{N}(d_0 - \sigma_d)$ vanishes faster than any exponential in d_0 while $C \sim e^{\sigma_d d_0 - \sigma_d^2/2}$, so $A/C \to 0$. ■